
Comparison methods for semilinear elliptic problems on Riemannian manifolds with a Ricci lower bound

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Abstract

In this article we develop a comparison method for positive solutions of the semilinear Dirichlet problem $\Delta u + f(u) = 0$ on a bounded domains $\Omega \Subset \mathcal{M}^n$ of a Riemannian manifold $(\mathcal{M}^n, g)$ with a Ricci lower bound $\text{Ric}_g \geq (n-1)kg$. Assuming admissibility and structural conditions on f , we prove a sharp pointwise gradient comparison, with a rigid characterization of the equality case. As applications, we derive an explicit isoperimetric-type inequality and a quantitative hot-spot localization estimate under natural convexity assumptions.

Key Words: Semilinear elliptic problem, gradient comparison, Ricci lower bound, hot spots.

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Contents

1	Introduction	1
1.1	Sketch of the construction of the comparison framework	3
1.2	Comparison results	6
2	Comparison method	8
2.1	Comparison framework	8
2.2	Model $\bar{r}$ -functions	12
2.3	Construction of the comparison framework	14
2.3.1	Existence and non-uniqueness of associated model pairs	16

3	Proof of the Comparison Results	18
3.1	Curvature estimates	20
3.2	An isoperimetric inequality	23
3.3	Location of the hot-spots	27
A	Appendix: ODE toolkit	29
B	Appendix: Gradient Estimates	32

1 Introduction

In elliptic PDE one often encounters a tight interplay between the geometry of a domain Ω and the analytic features encoded by a solution u . Global scales such as the inradius and the volume, together with local boundary information (for instance, bounds on principal curvatures or on the mean curvature of $\partial\Omega$), typically constrain the maximum value of u , the shape and location of its maximum set, and the behaviour of $|\nabla u|$ along level sets. In this article we will study these relations for semilinear elliptic problems on Riemannian manifolds under a lower bound on the Ricci tensor.

Let $(\mathcal{M}^n, g)$ be a complete Riemannian manifold that satisfies a lower bound on the Ricci tensor

$$\text{Ric}_g \geq (n - 1)k g \quad (1.1)$$

Let $\Omega \Subset \mathcal{M}$ be a bounded domain with $\mathcal{C}^2$ boundary and let $f \in \text{Lip}_{\text{loc}}(\mathbb{R})$. We consider

$$\begin{cases} \Delta u + f(u) = 0 & \text{in } \Omega, \\ u > 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \quad (1.2)$$

where Δ denotes the Laplace–Beltrami operator with respect to g . We will denote by $M := \max_{p \in \Omega} u(p)$ the maximum value of u and by $\text{Max}(u) := \{p \in \Omega : u(p) = M\}$ the maximum set.

We will show that, under suitable structural conditions on f and the lower bound (1.1), the norm of the gradient of u on each $\mathcal{U} \in \pi_0(\Omega \setminus \text{Max}(u))$ can be compared with the gradient of a one-dimensional model solution $\bar{U}$ defined on a warped product manifold, which is constructed from the data of (1.2) and the Ricci lower bound. Derived from this comparison, we obtain a curvature estimate on $\partial\Omega$ and $\text{Max}(u)$, an isoperimetric-type inequality, and a quantitative localization of hot spots.

The use of gradient estimates has been a widely studied topic with a variety of applications. One of the first was due to Pavel Kröger in 1992, who, in order to study the spectral gap, developed a method of model solutions on warped product manifolds for the Neumann eigenvalue problem on manifolds with Ricci curvature bounded below by a constant and with small spheres

as fibers, and then inverted that function to obtain a gradient estimate. Later on, Bakry and Qian in [?] extended this type of result to more general elliptic operators on $CD(k, n)$ spaces. This series of results on the spectral gap was subsequently extended to the p -Laplace operator by Valtorta in [?], and afterwards by Naber–Valtorta in the case $k < 0$ in [?].

In the case of gradient estimates for semilinear problems, it is worth mentioning the work of Farina and Valdinoci in $\mathbf{R}^n$ [?] and on Riemannian manifolds with a Ricci lower bound $k \geq 0$ in [?], in which they obtain Modica-type estimates for this family of problems with bounded solutions under certain assumptions on the mean curvature of the boundary. The proofs of these results rely on a P -function argument built from the solution u and its derivatives ∇u . Under the Dirichlet condition, they turn slope conditions into curvature conditions. Andrews and Xiong [?] developed a gradient comparison method in the spirit of Kröger, by inverting a model solution, for quasilinear problems on Riemannian manifolds with the Ricci lower bound of that paper [?, Theorem 27 and Theorem 30], and for viscosity solutions by means of the two-point function method. In both cases the mean curvature of the boundary is required to be nonnegative. By contrast, in our paper we trade the curvature condition on the boundary for the slope of the model solution at the boundary.

As we have already seen, there is a very close relation between the solutions and the geometry of the problem. Among other results, Agostiniani, Borghini and Mazzieri in [?] developed a comparison method for solutions of the torsion problem in the Euclidean plane, in which each solution is associated with a unique radial model solution; from this one obtains control on the norm of the gradient, which is then turned into a qualitative description of the solution, locating the set of maxima, in the spirit of Magnanini and Poggesi [?], as well as describing the curvature of certain level sets. In [?, Section 1.5] the authors discuss the possibility of extending their method to more general elliptic problems as we do in this article.

This philosophy was subsequently developed by the same authors in higher dimensions in Euclidean space [?], and extended to the 2-dimensional sphere by Espinar and Marín [?, ?] for more general problems, with applications to the critical catenoid problem. Similar comparison principles also arise in the analysis of static solutions of the Einstein equations with positive cosmological constant; see Borghini–Mazzieri and Borghini–Mazzieri–Chruściel [?, ?].

From [?, ?, ?, ?] we take the scheme: decompose Ω along $\text{Max}(u)$, associate with each region a radial model determined by the normalized slope at the wall, and invert the profile of the model in order to apply the maximum principle to an elliptic inequality with an explicit weight; from [?] we take the expansions at $\text{Max}(u)$. What changes is where the family of models comes from. In $\mathbf{R}^n$ and in $\mathbb{S}^2$ it is provided by the symmetry of the ambient space and rescaling makes it possible to normalize the maximum, so that a single parameter remains and the slope determines the model. With only a lower Ricci bound there is nothing with respect to which to be radial, and a general f on a curved background does not admit rescaling: the family must be constructed out of Ω itself, over a component of its boundary, with two parameters, and the slope no longer determines the model, but merely bounds it.

Compared with the model solutions of [?, ?], where the comparison family has only one

parameter, the generalization to Riemannian manifolds and to more general problems makes both the family and its behavior more involved. In [?] the slope at the wall serves to *guess* the core radius, and one compares with the model having exactly that slope; this is possible there because the functions $\bar{r}^\pm$ are monotone and bijective. In our setting, for a general f they are not (they are if $f \geq x f'$, see Proposition 2.8) and, moreover, there are slopes that no model attains.

Neither of the two lines of research therefore covers problem (1.2) under the Ricci lower bound. The first needs the symmetry of the ambient space in order to have a comparison family, and rescaling in order to fix the scale; the second replaces the boundary datum by a curvature condition, works with nondegenerate models, and its equality case is exceptional. What is missing is a comparison family that depends neither on symmetry nor on scale, that requires no curvature conditions at the boundary, and that admits an equality case; constructing it out of Ω is what this paper does, and it is what yields the three theorems that follow.

1.1 Sketch of the construction of the comparison framework

To obtain this result we have developed a comparison method that extends the ideas of [?, ?, ?, ?] to semilinear elliptic problems in a Riemannian setting under a Ricci lower bound. This enhanced comparison scheme is based on model solutions, i.e. solutions of problem (1.2) on warped product manifolds whose base is an interval $I_k := [0, \bar{r}_k]$ and whose fiber is a connected component of the boundary of Ω , $\Gamma_0 \in \pi_0(\partial\Omega)$, equipped with the induced metric g_{Γ_0} ,

$$(\mathcal{M}_k^n(\Gamma_0), g_k) = (I_k \times \Gamma_0, dr^2 + s_k^2(r) g_{\Gamma_0}).$$

As for the warping function, depending on the sign of k , it will be given by

$$s_k(r) = \begin{cases} \sin(\sqrt{k} r)/\sqrt{k} & \text{if } k > 0, \\ r & \text{if } k = 0, \\ \sinh(\sqrt{-k} r)/\sqrt{-k} & \text{if } k < 0. \end{cases}$$

In this way, when Γ_0 is a sphere, as for instance in the works of Kröger[?] or Naber–Valtorta [?], the model manifolds are the space forms endowed with the metric with a pole.

The model solutions will be radially symmetric, that is, constant on each fiber, so that problem (1.2) reduces to the Cauchy problem

$$\begin{cases} U''(r) + (n-1) \cot_k(r) U'(r) + f(U(r)) = 0, \\ U(R) = M, \quad U'(R) = 0, \end{cases}$$

where $\cot_k(r) = s'_k(r)/s_k(r)$. Here $M > 0$ is the maximum value of the profile and $R \in I_k$ is the radius at which it is attained, which we call the **core radius**. Once k and f are fixed, the pair

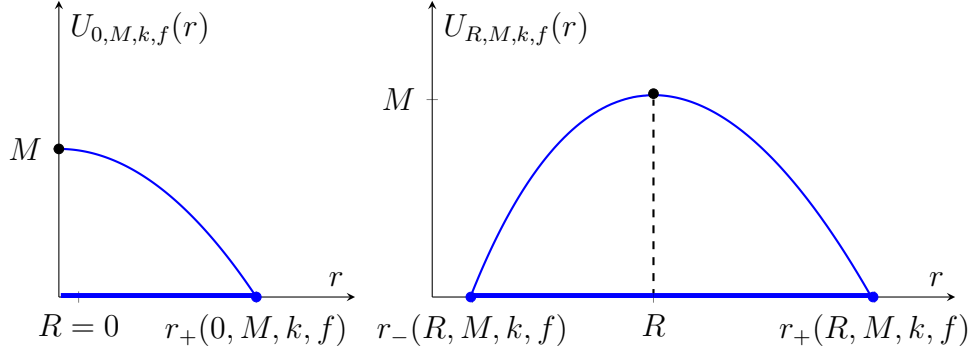


Figure 1: Here we show the profile of a solution $U_{R,M,k,f}$ to (2.3) when f is k -admissible. *Left:* case $R = 0$, where the solution $U_{0,M,k,f}$ is positive on $(0, r_+(0, M, k, f))$, vanishes at $r = r_+(0, M, k, f)$, and attains its maximum at the origin. *Right:* case $R > 0$, where $U_{R,M,k,f}$ is supported on the interval $(r_-(R, M, k, f), r_+(R, M, k, f))$, vanishes at both boundary radii, and attains its maximum at $r = R$.

(R, M) is the only remaining freedom, and every object built out of it carries it as a subscript: we write $U_{R,M}$ for the profile, $r_{\pm}(R, M)$ for the two radii at which it vanishes, and

$$\Omega_{R,M} \in \mathcal{M}_k^n(\Gamma_0), \quad u_{R,M} := U_{R,M} \circ r,$$

for the domain it sweeps out — of ball type if $R = 0$, of shell type if $R > 0$ — together with the solution it carries. The profile itself does not depend on the fiber; only the domain does, and the level sets $\{r = \text{const}\}$ are homothetic to (Γ_0, g_{Γ_0}) . Throughout the introduction we suppress the fixed data (k, f) ; the full notation $U_{R,M,k,f}$, $\Omega_{R,M,k,f}$, $r_{\pm}(R, M, k, f)$ is the one used in Section 2.

Not every nonlinearity produces profiles of this shape: we need them to be positive between their extreme zeros, to vanish there, and to be strictly monotone on either side of $r = R$; see Figure 1. We encode this in the notion of k -admissibility (Definition 2.3), which holds in particular whenever $f > 0$ and $f(x) \geq nkx + f(0)$ (Proposition 2.4).

It remains to decide which model corresponds to each region. We measure the steepness of a model at the wall by comparing it with that of the solution on the disk with the same maximum,

$$\bar{\tau}_{M,k,f}^{\pm}(R) = \frac{U'_{R,M}(r_{\pm}(R))^2}{U'_{0,M}(r_+(0))^2}, \quad \text{and} \quad \bar{\tau}(\mathcal{U}) = \frac{\max_{\partial\Omega \cap \text{cl}(\mathcal{U})} |\nabla u|^2}{U'_{0,M}(r_+(0))^2},$$

ratios which do not depend on the fiber.

Therefore, to be able to perform the comparison we will define the notion of *comparison pair* that will be a model solution $(\bar{\mathcal{U}}, \bar{U})$ such that $\bar{\tau}(\mathcal{U}) \leq \bar{\tau}_{M,k,f}^{\pm}(R)$. In the equality case we will name it as *associated model pair*. The existence of the associated model pair is not always guaranteed, but it is ensured to have a comparison pair, as we will see in Lemma 2.14.

Once we fix a comparison pair $(\bar{\mathcal{U}}, \bar{u})$ for $(\mathcal{U}, u)$, write $\bar{U}$ for its radial profile, $\bar{R}$ for its core radius and $\bar{r}_\pm = r_\pm(\bar{R}, M, k, f)$ for its zeros. The model carries a radial coordinate and $\mathcal{U}$ does not, so in order to compare the two we transport that coordinate to $\mathcal{U}$ through u itself. This is where k -admissibility enters: it guarantees that $\bar{U}$ is positive between its zeros and strictly monotone on each side of the core, hence invertible on each branch. Denoting by

$$\chi^\diamond: [0, M] \longrightarrow \begin{cases} [\bar{r}_-, \bar{R}], & \diamond = -, \\ [\bar{R}, \bar{r}_+], & \diamond = +, \end{cases}$$

the inverse of $\bar{U}$ on the branch singled out by the comparison pair, we call

$$\Psi := \chi^\diamond \circ u: \mathcal{U} \longrightarrow [\bar{r}_-, \bar{r}_+]$$

the *pseudo-radial function* of $(\mathcal{U}, u)$ relative to $(\bar{\mathcal{U}}, \bar{u})$. It captures, at each point p , the radius at which the model attains the value $u(p)$: the wall $\partial\Omega \cap \text{cl}(\mathcal{U})$ is sent to the endpoint $\bar{r}_\diamond$, the set $\text{Max}(u)$ to the core $\bar{R}$, and on the model itself Ψ is exactly the radial coordinate.

In [?, §3.1] and [?, §4] the analogous function is obtained by applying the implicit function theorem to an explicit expression for the model profile. No such expression is available here, and the inversion is performed on the solution of (??) itself; this is why monotonicity has to be built into the class of nonlinearities rather than verified case by case.

1.2 Comparison results

Let (Ω, u) solve the semilinear Dirichlet problem (1.2) on an n -dimensional Riemannian manifold $(\mathcal{M}, g)$ satisfying the Ricci lower bound (1.1), and let $f \in \text{Lip}_{\text{loc}}(\mathbb{R})$ be k -admissible in the sense of Definition 2.3. Fix a component

$$\mathcal{U} \in \pi_0(\Omega \setminus \text{Max}(u)) \quad \text{with} \quad M := u_{\max} \in \mathcal{I}_f,$$

and fix the model manifold constructed in the introduction, namely $(\mathcal{M}_k^n(\Gamma_0), g_k)$. Let $(\bar{\mathcal{U}}, \bar{u})$ be a comparison pair for $(\mathcal{U}, u)$ in the sense of Definition 2.11. From now on a bar marks the model side of the comparison: $(\bar{\mathcal{U}}, \bar{u})$ is the comparison pair, $\bar{U}$ its profile, $\bar{R}$ its core radius, $\bar{r}_\pm := r_\pm(\bar{R}, M)$ the radii where $\bar{U}$ vanishes, and $\bar{\Gamma}_M$ its maximum set. Let Ψ be the associated pseudo-radial function (Definition 2.12), so that $u = \bar{U} \circ \Psi$ on $\mathcal{U}$.

Theorem A (Gradient estimates). *Let $f \in \mathcal{C}^1(\mathbb{R})$ be k -admissible and assume that*

$$f'(x) \geq nk \quad \text{for all } x \in \mathcal{I}_f. \tag{1.3}$$

Then, for any comparison pair $(\bar{\mathcal{U}}, \bar{u})$ for $(\mathcal{U}, u)$,

$$|\nabla u|^2(p) \leq \bar{U}'(\Psi(p))^2 \quad \text{for all } p \in \mathcal{U}.$$

Moreover, if equality holds at one point of $\mathcal{U}$, then $\text{Ric}(\nabla u, \nabla u) = (n-1)k|\nabla u|^2$ on $\mathcal{U}$, the metric g restricted to $\mathcal{U}$ is isometric to that of the model manifold $(\mathcal{M}_k^n(\Gamma_0), g_k)$, and $(\mathcal{U}, u)$ is congruent to $(\bar{\mathcal{U}}, \bar{u})$.

Remark 1.1. Since $u = \bar{U} \circ \Psi$, the conclusion reads $|\nabla \Psi| \leq 1$: the pseudo-radial function is 1-Lipschitz, so the level sets of u in $\mathcal{U}$ are at least as far apart as the corresponding level sets of the model. This is the form in which the comparison appears in [?, ?].

Condition (2.13) enters the proof at a single point. The comparison rests on an elliptic inequality satisfied by the difference $|\nabla u|^2 - \bar{U}'(\Psi)^2$, weighted so that it extends continuously to $\text{Max}(u)$ (Lemma B.1). That difference is non-positive on $\partial\mathcal{U}$ by the very definition of a comparison pair, and (2.13) is what gives the zeroth order coefficient of the inequality the sign required for the maximum principle to carry this into $\mathcal{U}$. The condition is sufficient for that, and it is not necessary; we discuss the gap in Subsection 3.

As consequences of Theorem A, we obtain curvature bounds for $\partial\Omega$ (Proposition 3.2) and for the top level set (Proposition 3.4), as well as an area bound for the top level set (Proposition 3.5). In higher dimensions, these curvature estimates are naturally weaker than in the surface case: since the PDE controls Δu the comparison yields effective control of the *mean* curvature of level sets but not of the full principal curvature spectrum when $n \geq 3$ (see [?]).

Combining the area bound with the coarea formula yields the following isoperimetric-type inequality. For any $s > 0$, denote by $\mathcal{H}^s$ the s -dimensional Hausdorff measure associated to the metric g of $\mathcal{M}$, and by $\mathcal{H}_k^s$ that associated to the metric g_k of the model manifold $\mathcal{M}_k^n(\Gamma_0)$.

The proof of this result is given in Section 3.2.

Theorem B (Isoperimetric inequality). *Let $f \in C^m(\mathbb{R})$, $m = \max\{n-2, 3\}$, be k -admissible and satisfy (2.13). Assume that the $(n-1)$ -dimensional part of $\text{cl}(\mathcal{U}) \cap \text{Max}(u)$, denoted by Γ_M , is a (possibly disconnected) Lipschitz hypersurface. Then, for any comparison pair $(\bar{\mathcal{U}}, \bar{u})$ with positive core radius,*

$$\frac{\mathcal{H}^n(\mathcal{U})}{\mathcal{H}^{n-1}(\Gamma_M)} \geq \begin{cases} \frac{\mathcal{H}_k^n(\mathcal{U}_{\bar{R}, M, k, f}^+)}{\mathcal{H}_k^{n-1}(\Gamma_{\bar{R}, M, k, f})} & \text{if } \bar{\mathcal{U}} \in \text{Model}_{M, k, f}^+ \\ \frac{\mathcal{H}_k^n(\mathcal{U}_{\bar{R}, M, k, f}^-)}{\mathcal{H}_k^{n-1}(\Gamma_{\bar{R}, M, k, f})} & \text{if } \bar{\mathcal{U}} \in \text{Model}_{M, k, f}^- \end{cases} \quad (1.4)$$

Furthermore, equality holds in (1.4) if, and only if $(\mathcal{U}, u) \equiv (\bar{\mathcal{U}}, \bar{u})$.

We conclude with a quantitative localization estimate for the maximum set (hot spots), extending the sharp bounds obtained for Serrin-type equations in [?, ?]. The proof is given in Section 3.3.

Theorem C (Location of the hot-spots). *Let $f \in C^1(\mathbb{R})$ be a k -admissible function satisfying the condition (2.13) and*

$$f(x) \leq f_k(x) = nkx + 1$$

Assume that Ω is contained in a convex domain of $\mathcal{M}$ and that

$$\max_{p \in \partial\Omega \cap \partial\mathcal{U}} |\nabla u|^2(p) = \max_{q \in \partial\Omega} |\nabla u|^2(q).$$

If $k > 0$ suppose that $\bar{r}_+ \leq \bar{r}_k/2$. Let $(\bar{\mathcal{U}}, \bar{u})$ be an comparison pair to $(\mathcal{U}, u)$ with core radius $\bar{R}$. Then,

$$\frac{\text{dist}(p, \partial\Omega)}{\tan_k(r_\Omega)/n} \begin{cases} \geq \frac{\bar{r}_+ - \bar{R}}{\sqrt{\frac{2}{n}M + kM^2}} & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+, \\ > \frac{\bar{R} - \bar{r}_-}{\sqrt{\frac{2}{n}M + kM^2}} & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^-, \end{cases} \quad \text{for all } p \in \text{Max}(u). \quad (1.5)$$

where $r_\Omega = \max \{ \text{dist}(p, \partial\Omega) : p \in \Omega \}$. Moreover, if equality holds at a point of Ω , then $(\mathcal{M}, g)$ is isometric to the space form $\mathbb{M}^n(k)$, $f(x) = nkx + 1$, and (Ω, u) is a ball solution of Serrin's problem.

The additional hypotheses in Theorem C have distinct roles. The upper bound $f(x) \leq nkx + 1$ ensures that the Serrin's ball profile provides a subsolution, yielding a lower bound for u in terms of the distance to $\partial\Omega$ by the maximum principle. Condition (2.13) (i.e. $f'(x) \geq nk$) is used for the gradient comparison in Theorem A. Moreover, the condition over $\mathcal{U}$, $\bar{\tau}(\mathcal{U}) = \bar{\tau}(\Omega)$ is to be able to apply Lemma 3.7 to obtain a lower bound for $u(p)$.

Organisation of the paper. In Section 2 we construct the comparison framework. Section 3 contains the proofs of these comparison results. Appendix A collects the ODE statements used in Section 2 and Appendix B proves an elliptic inequality needed for the gradient estimates.

2 Comparison method

In this section, we set up the comparison framework, extending the strategy of [?, ?] to solutions of (1.2) under the Ricci lower bound (1.1). We work with boundary-adapted model manifolds realized as warped products and with the associated model solutions by reducing (1.2) to a radial ODE for fiber-invariant functions and then lifting the resulting profiles to canonical radial domains (of disk- and annulus-type). A normalized boundary-response parameter (the $\bar{\tau}$ -functions) then selects comparison and associated model pairs for each connected component of $\Omega \setminus \text{Max}(u)$, leading to the gradient and geometric estimates stated in Subsection 1.2.

2.1 Comparison framework

Unlike the planar ring-shaped setting, where the reference family is fixed a priori, here the comparison family must be built from the data of (1.2). We do so by introducing boundary-adapted *model manifolds*, realized as warped products. This class includes the space forms and allows curvature information to be encoded through the warping function; it has been used, for instance, in eigenvalue comparisons on geodesic balls [?, ?, ?]. In these Riemannian manifold structure the Laplace–Beltrami operator takes a particularly simple form, which we will use repeatedly; see also [?, ?, ?, ?, ?] for existence and regularity results for semilinear problems on related models.

Definition 2.1 (Model Manifold). *Given $k \in \mathbb{R}$, we say that a Riemannian manifold $(\mathcal{M}^n, g)$ is a model manifold if it is a warped product of the form:*

$$(\mathcal{M}_k^n(F), g_k) := (I_k \times F, dr^2 + s_k(r)^2 g_F),$$

where the fiber F is an $(n - 1)$ –dimensional smooth compact Riemannian manifold and $I_k := [0, \bar{r}_k)$, where $\bar{r}_k := +\infty$ if $k \leq 0$ and $\bar{r}_k := \pi/\sqrt{k}$ if $k > 0$. The warping function is defined as,

$$s_k(r) := \begin{cases} \frac{\sinh(\sqrt{-k}r)}{\sqrt{-k}} & \text{if } k < 0 \\ r & \text{if } k = 0, \\ \frac{\sin(\sqrt{k}r)}{\sqrt{k}} & \text{if } k > 0, \end{cases} \quad (2.1)$$

Remark 2.2 (Conical singularity when $(F, g_F) \not\cong \mathbb{S}^{n-1}$). *If $(F, g_F) \not\cong (\mathbb{S}^{n-1}, g_{\text{round}})$, the warped product metric $g_k = dr^2 + s_k(r)^2 g_F$ does not extend smoothly across $r = 0$. Indeed, since $s_k(r) = r + O(r^3)$ as $r \rightarrow 0$, a smooth completion by adding a pole o would force g_k to be asymptotically Euclidean in geodesic polar coordinates, which is equivalent to the link metric g_F being the unit round metric. In general, the metric completion has an isolated conical singularity at $r = 0$, locally modeled on the cone $(0, \varepsilon) \times F$ with metric $dr^2 + r^2 g_F$; see [?].*

For later use we introduce the following auxiliary notation,

$$\cot_k(r) := \frac{s'_k(r)}{s_k(r)} \quad \forall r \in (0, \bar{r}_k) \quad \text{and} \quad \tan_k(r) = \frac{s_k(r)}{s'_k(r)}, \quad \forall r \in [0, \bar{r}_k) \setminus \{\bar{r}_k/2\}.$$

Since the metric has a warped product structure, the Laplace–Beltrami operator on a model manifold takes the following form,

$$\Delta = \partial_r^2 + (n - 1) \cot_k(r) \partial_r + \frac{1}{s_k(r)^2} \Delta_F, \quad (2.2)$$

Having fixed the class of model manifolds, we now introduce the corresponding model solutions. In the space–form case (that is, when $F = \mathbb{S}^{n-1}$ with the round metric up to scaling),

the natural reference class consists of radial solutions about a point o , namely functions of the distance $r(p) = \text{dist}(o, p)$. More generally, on a model manifold $(\mathcal{M}_k^n(F), g_k)$ with $n \geq 2$, we call *radial* those solutions of (1.2) that depend only on the base coordinate r .

A direct computation using (2.2) shows that if $u(p) = U(r(p))$, with $U \in \mathcal{C}^2(\mathbb{R})$, then u solves $\Delta u + f(u) = 0$ if and only if U solves the ordinary differential equation

$$U''(r) + (n-1) \cot_k(r) U'(r) + f(U(r)) = 0, \quad (2.3)$$

for $r \in (0, \bar{r}_k)$, where $\bar{r}_k := \pi/\sqrt{k}$ if $k > 0$ and $\bar{r}_k := +\infty$ otherwise. Thus, we can find solutions U of (2.3) satisfying the Cauchy conditions

$$U(R) = M \quad \text{and} \quad U'(R) = 0, \quad (2.4)$$

for some $R \in [0, \bar{r}_k)$ and $M > 0$. The parameter R will be referred to as the *core radius*. For $R > 0$, the maximum of u is attained on the hypersurface $\{r = R\}$, whereas for $R = 0$ the maximum is attained at the origin (the *centered* case).

For the comparison we need profiles that stay positive between their first and last zeros, are strictly monotone away from $r = R$, and vanish at the boundary radii of the associated model domain, see Figure 1. Since this is not automatic for a general locally Lipschitz f , we encode it in the notion of k -admissibility; see Definition 2.3.

Definition 2.3 (Admissible nonlinearities). *Given $k \in \mathbb{R}$, a locally Lipschitz function $f \in \text{Lip}_{\text{loc}}(\mathbb{R})$ is called k -admissible if there exist non-empty connected intervals*

$$0 \in \mathcal{R}_f \subset [0, \bar{r}_k) \quad \text{and} \quad \mathcal{I}_f \subset \mathbb{R}_+ \text{ with } 0 \in \partial \mathcal{I}_f$$

such that, for every $(R, M) \in \mathcal{R}_f \times \mathcal{I}_f$, there exists a maximal solution $U_{R,M,k,f}$ of (2.3)–(2.4) that is defined on an interval that contains a closed interval $[r_-(R, M, k, f), r_+(R, M, k, f)]$ with

$$0 \leq r_-(R, M, k, f) \leq R < r_+(R, M, k, f) \leq \bar{r}_k,$$

and satisfies:

- *If $R = 0$, then $r_-(0, M, k, f) = 0 < r_+(0, M, k, f)$,*
 - $U_{0,M,k,f} > 0$ on $[0, r_+(0, M, k, f))$,
 - $U'_{0,M,k,f} \neq 0$ on $(0, r_+(0, M, k, f))$, and
 - $U_{0,M,k,f}(r_+(0, M, k, f)) = 0$.
- *If $R > 0$, then $0 < r_-(R, M, k, f) < R < r_+(R, M, k, f)$,*
 - $U_{R,M,k,f} > 0$ on $(r_-(R, M, k, f), r_+(R, M, k, f))$,

- $U'_{R,M,k,f} \neq 0$ on $(r_-(R, M, k, f), r_+(R, M, k, f)) \setminus \{R\}$, and
- $U_{R,M,k,f}(r_\pm(R, M, k, f)) = 0$.

The family of k -admissible functions comprises a wide class of nonlinearities, for example the Helmholtz equation, studied in [?], and its affine version; the Lane–Emden equation $f(x) = x|x|^p$; or Allen–Cahn–type terms of the form $f(x) = x - x^p$, where $p \in (1, \frac{n+2}{n-2})$ in $\mathbb{S}^d$, for $n > 1$, as studied in [?].

When $k > 0$, the solutions to (2.3)-(2.4) enjoys the following symmetry: if $U_{R,M,k,f}(r)$ is a solution with core radius R , then the solution with core radius $\bar{r}_k - R$ satisfies that $U_{R,M,k,f}(r) = U_{\bar{r}_k - R, M, k, f}(\bar{r}_k - r)$. Thus, the solution whose core radius is $R = \bar{r}_k/2$ is symmetric with respect to the line $r = R$.

Note that k -admissibility forces $f > 0$ on $\mathcal{I}_f$. Otherwise, if there exists $M_0 \in \mathcal{I}_f$ with $f(M_0) \leq 0$, then (2.3)-(2.4) would imply that all functions $U_{R,M_0,k,f}$ have either a minimum or a saddle point at $r = R$, contradicting the properties stated above.

In general, it is not clear what the most general conditions on f are that ensure k -admissibility for the problem (2.3)-(2.4). We will be able to guarantee k -admissibility of a locally Lipschitz function $f \in \text{Lip}_{loc}(\mathbb{R})$ under the following conditions:

Standard Conditions.

There exists an interval $I_f \subset \mathbb{R}_+$ with $0 \in \partial I_f$ such that

$$f(x) > 0 \quad \text{and} \quad f(x) \geq nkx + f(0), \quad \forall x \in I_f,$$

where $f(0) > 0$ if $k \leq 0$.

The Standard Conditions take a particularly simple form when $k = 0$, since they reduce to $f(x) \geq f(0)$ on I_f , so f is bounded from below by its value at the origin. Indeed, the function $f(x) \equiv 0$ is not 0-admissible. When $k < 0$, the term nkx is non-positive for $x \geq 0$, hence the constraint $f(x) \geq nkx + f(0)$ is weaker than $f(x) \geq f(0)$; in particular, it does *not* force f to be non-decreasing near 0, but it still requires $f(x) > 0$ on I_f . In fact, the function $f(x) = nkx$ is not k -admissible when $k < 0$.

The case $k > 0$ (for instance on spherical geometries) is more restrictive, because nkx is increasing and the condition imposes a definite lower growth in x . For example, for an affine nonlinearity $f(u) = \lambda u + \beta$ (an affine Helmholtz-type term), the inequality $f(x) \geq nkx + f(0)$ is equivalent to $\lambda \geq nk$ (since $f(0) = \beta$), together with $f(x) > 0$ on I_f . Another standard example is the Liouville–Bratu–Gelfand–type nonlinearity $f(u) = Ke^{2u}$ with $K \in \mathbb{R}$, so the standard condition holds provided $K \geq nk$. The study of this problem in dimension 2 is linked to prescribing curvatures in the disk via conformal changes; see, for example, [?, ?].

For power-type nonlinearities, one should keep in mind the positivity requirement $f(x) > 0$ on I_f . For instance, $f(x) = x - x^p$ is non-negative for $x \in [0, 1]$, hence it satisfies the standard

conditions on an interval with 0 in its boundary. Similarly, for Brezis–Nirenberg-type terms of the form $f(x) = x^{2^*-1} + \lambda x$ (with $2^* = \frac{2n}{n-2}$ for $n \geq 3$), the inequality near 0 forces $\lambda \geq nk$.

The following result is proved in Appendix ??.

Proposition 2.4. *Let $k \in \mathbb{R}$ and $f \in \text{Lip}_{\text{loc}}(\mathbb{R})$ satisfying the Standard Conditions. Then f is k –admissible with admissible sets $\mathcal{I}_f = I_f$ and $\mathcal{R}_f = [0, \bar{r}_k)$.*

Remark 2.5. *The previous proposition remains true if the Standard Conditions are replaced by a more technical assumption: namely, that $f \geq \tilde{f}$ on I_f , where $\tilde{f}$ admits a solution to (2.3) with maximum value $M > 0$ on a bounded interval, for every $M \in I_f$. This can be seen by inspecting the proof in Appendix ??.*

Given a model manifold $(\mathcal{M}_k^n(F), g_k)$, assume $f \in \text{Lip}_{\text{loc}}(\mathbb{R})$ is a k –admissible function. Let $U_{R,M,k,f}$ be the solution of (2.3)–(2.4) constructed above. The profile of $U_{R,M,k,f}$ depends on the geometric data, the dimension n , the curvature bound k , the nonlinearity f , and on the Cauchy data, (R, M) , but not on the fiber F . For each $(R, M) \in \mathcal{R}_f \times \mathcal{I}_f$, define the domain

$$\Omega_{R,M,k,f} := \begin{cases} \{p \in \mathcal{M}_k^n(F) : r(p) < r_+(0, M, k, f)\}, & R = 0, \\ \{p \in \mathcal{M}_k^n(F) : r_-(R, M, k, f) < r(p) < r_+(R, M, k, f)\}, & R \neq 0. \end{cases}$$

and set

$$u_{R,M,k,f}(p) := U_{R,M,k,f}(r(p)), \quad p \in \Omega_{R,M,k,f}.$$

Then $(\Omega_{R,M,k,f}, u_{R,M,k,f})$ solves (1.2) and on each boundary component of $\Omega_{R,M,k,f}$ the norm of the gradient of $u_{R,M,k,f}$ is locally constant on each boundary component.

The pairs $(\Omega_{R,M,k,f}, u_{R,M,k,f})$ comprise all positive radial solutions of (1.2) with constant Neumann datum in each connected component of the model manifold and will be the reference objects for comparison. Moreover:

- If $R = 0$, $\Omega_{0,M,k,f}$ is called the *centered domain*, and we set

$$\Gamma_{0,M,k,f} = u_{0,M,k,f}^{-1}(M) \text{ and } \Gamma_{0,M,k,f}^+ = \{p \in \mathcal{M}_k^n(F) : r(p) = r_+(0, M, k, f)\}.$$

We also denote by $\mathcal{U}_{0,M,k,f}^+ \subset \Omega_{0,M,k,f} \setminus \Gamma_{0,M,k,f}$ the subdomain of $\Omega_{0,M,k,f}$ without maximum points and we set $\mathcal{U}_{0,M,k,f}^- = \emptyset$. The boundary of $\mathcal{U}_{0,M,k,f}^+$ is connected if F is a connected manifold.

- If $R \in (0, \bar{r}_k)$, we denote

$$\Gamma_{R,M,k,f} = u_{R,M,k,f}^{-1}(M) \text{ and } \Gamma_{R,M,k,f}^\pm = \{p \in \mathcal{M}_k^n(F) : r(p) = r_\pm(R, M, k, f)\},$$

and $\mathcal{U}_{R,M,k,f}^\pm \subset \Omega_{R,M,k,f} \setminus \Gamma_{R,M,k,f}$ the subdomains of $\Omega_{R,M,k,f}$ without maximum points such that $\text{cl}(\mathcal{U}_{R,M,k,f}^\pm) \cap \partial\Omega_{R,M,k,f} = \Gamma_{R,M,k,f}^\pm$.

Definition 2.6 (Model pairs). *Given a model manifold $(\mathcal{M}_k^n(F), g_k)$, let $f \in \text{Lip}_{loc}(\mathbb{R})$ be a k -admissible function. For $(R, M) \in \mathcal{R}_f \times \mathcal{I}_f$, define the sets $\mathcal{U}_{R,M,k,f}^\pm$ as above and the corresponding functions*

$$u_{R,M,k,f}^\pm(p) := U_{R,M,k,f}(r(p)) \quad \text{on } \mathcal{U}_{R,M,k,f}^\pm.$$

Then, the pairs $(u_{R,M,k,f}^\pm, u_{R,M,k,f}^\pm)$ are called model pairs and we denote by $\text{Model}_{M,k,f}^\pm$ the family of model pairs $(u_{R,M,k,f}^\pm, u_{R,M,k,f}^\pm)$ obtained by varying the core radius $R \in \mathcal{R}_f$ of the solution.

We call $(u_{0,M,k,f}^+, u_{0,M,k,f}^+)$ the centered model pair of $\text{Model}_{M,k,f}^+$. Also, we write $\text{Model}_{M,k,f} = \text{Model}_{M,k,f}^+ \sqcup \text{Model}_{M,k,f}^-$ for the complete family of model pairs.

2.2 Model $\bar{\tau}$ -functions

To track how the choice of initial data affects the boundary behavior of these solutions, we introduce the $\bar{\tau}$ -functions. These functions encode the boundary response of the radial profiles and provide a convenient way to compare their boundary behavior with that of the centered solution.

Given a model manifold $(\mathcal{M}_k^n(F), g_k)$, let $f \in \text{Lip}_{loc}(\mathbb{R})$ be a k -admissible function. Since $f \circ u_{0,M,k,f}^+$ is positive in $\mathcal{U}_{0,M,k,f}^+$ for each $M \in \mathcal{I}_f$, Hopf's lemma yields that the centered model solution $(u_{0,M,k,f}^+, u_{0,M,k,f}^+)$ determines a positive constant

$$c(M, k, f) := |\nabla u_{0,M,k,f}^+|^2|_{\Gamma_{0,M,k,f}^+} = U'_{0,M,k,f}(r_+(0, M, k, f))^2 > 0, \quad (2.6)$$

which depends only on (M, k, f) (and not on the choice of fiber F). We define the *model $\bar{\tau}$ -functions* by

$$\bar{\tau}_{M,k,f}^\pm(R) := \frac{U'_{R,M,k,f}(r_\pm(R, M, k, f))^2}{U'_{0,M,k,f}(r_+(0, M, k, f))^2} = \frac{|\nabla u_{R,M,k,f}^\pm|^2|_{\Gamma_{R,M,k,f}^\pm}}{c(M, k, f)}, \quad R \in [0, \bar{r}_k). \quad (2.7)$$

These ratios encode how the boundary gradient of a model solution compares to that of the centered solution, and they will be used to parametrize comparison pairs. For us it will be useful to introduce the following notation for the lower and upper bounds of the functions $\bar{\tau}_{M,k,f}^\pm$,

$$\tau_{k,f}^-(M) := \inf_{R \in [0, \bar{r}_k)} \bar{\tau}_{M,k,f}^-(R), \quad \tau_{k,f}^+(M) := \sup_{R \in [0, \bar{r}_k)} \bar{\tau}_{M,k,f}^+(R)$$

and

$$\tau_{k,f}^0(M) := \inf_{R \in [0, \bar{r}_k)} \bar{\tau}_{M,k,f}^+(R).$$

In [?, ?, ?], the monotonicity of the $\bar{\tau}$ -functions is clear due to the nature of the nonlinearity. However, in the present setting the monotonicity of the model $\bar{\tau}$ -functions is not evident, and even determining the range of their images is not immediate. Thanks to the maximum principle we can state some results under some restrictions on the nonlinearity.

In the special case $f(x) = f_k(x) := nkx + 1$, the problem (2.3)–(2.4) corresponds to Serrin's equation. In this case, it is easy to check that $\bar{\tau}_{M,k,f}^-(R) \rightarrow +\infty$ as $R \rightarrow 0^+$. We prove it in Proposition A.1, in Appendix ???. In this appendix, we also provide the proof of the following result:

Proposition 2.7. *Let f be a function satisfying the Standard Conditions. Then, for any $M \in \mathcal{I}_f$, $\bar{\tau}_{M,k,f}^-(R) \rightarrow +\infty$ as $R \rightarrow 0^+$. Moreover, one has $\tau_{k,f}^+(M) \leq \tau_{k,f}^-(M) < +\infty$ if $k \leq 0$, and $\tau_{k,f}^+(M) \geq \tau_{k,f}^-(M)$ if $k > 0$.*

Finally, note that under an additional condition on f one can show monotonicity of $\bar{\tau}_{M,k,f}^\pm$. Again, we postpone the proof of this result to Appendix ??.

Proposition 2.8. *Let $f \in \mathcal{C}^1(\mathbb{R})$ satisfy the Standard Conditions, and suppose there exists $\bar{M} > 0$ such that $(0, \bar{M}) \subset \mathcal{I}_f$ and*

$$f(x) \geq xf'(x) \quad \text{for all } 0 < x < \bar{M}. \quad (2.8)$$

Then for any $M \in (0, \bar{M})$ we have $\tau_{k,f}^0(M) = \bar{\tau}_{M,k,f}^+(0) = 1$, and $\bar{\tau}_{M,k,f}^-(R)$ is strictly decreasing while $\bar{\tau}_{M,k,f}^+(R)$ is strictly increasing on $(0, \bar{r}_k)$.

For future use, we define the *admissible set* and *gap* associated to (M, k, f) as

$$\text{Adm}_{M,k,f} := \text{Im}(\bar{\tau}_{M,k,f}^-) \cup \text{Im}(\bar{\tau}_{M,k,f}^+), \quad \text{Gap}_{M,k,f} := \mathbb{R}_+^* \setminus (\text{Adm}_{M,k,f} \cup [0, \tau_{k,f}^0(M)]). \quad (2.9)$$

Note that, by the continuous dependence with respect to the initial data of solutions to differential equations with locally Lipschitz coefficients (see for example [?, Theorem 2.21]), it follows that $\text{Im}(\bar{\tau}_{M,k,f}^-)$ and $\text{Im}(\bar{\tau}_{M,k,f}^+)$ are connected intervals. Therefore, $\text{Adm}_{M,k,f}$ is an interval or the union of two disjoint intervals, and then $\text{Gap}_{M,k,f}$ can be empty, a point or the union of intervals with non-empty interior. As we shall see later, these three options can actually occur (see Remark 2.9 and Corollary 2.14).

Proposition 2.7 implies that, when f satisfies the Standard Conditions, $\text{Gap}_{M,k,f} = \emptyset$ when $k > 0$ and consists of a bounded interval with extrema $\tau_{k,f}^+(M)$ and $\tau_{k,f}^-(M)$ if $k \leq 0$. Here, we consider that $[a, a] = \{a\}$ for any $a \in \mathbb{R}$.

Remark 2.9. *Proposition 2.8 and the symmetry of the solutions to (2.3)–(2.4) when $k > 0$ imply that, when f satisfies condition (2.8) and Standard Conditions, we get that*

$$\tau_{k,f}^\pm(M) = \lim_{R \rightarrow \bar{r}_k/2} \bar{\tau}_{M,k,f}^\pm(R), \quad \forall M \in \mathcal{I}_f. \quad (2.10)$$

When $k = 0$ and f is the Serrin's problem non-linearity this result also holds. This is already noted in [?, ?], and it also holds that $\bar{\tau}_{0,f_0}^+(M) = \bar{\tau}_{0,f_0}^-(M)$. This implies that $\text{Gap}_{M,0,f_0}$ consists of a single value, $\{\bar{\tau}_{0,f_0}^+(M)\}$. However, when $k < 0$ using (2.10) we can show that Gap_{M,k,f_k} consists of a closed interval with non-empty interior.

In fact, given any $M \in (0, 1/n)$, write $V_{R,M,-1}(r)$ for the solution to (2.3)-(2.4) with $k = -1$ and $f(x) = f_{-1}(x) = -nx + 1$. Then, by doing the change of variable $r = R + s$, it is straightforward to show that $V_{R,M,-1}$ converges (to first order) to the function

$$V_{\infty,M}(s) = 1 - \frac{1}{(n+1)(M+1)} (e^{-ns} + n e^s)$$

as $R \rightarrow +\infty$. Thus, if $s_- < 0 < s_+$ denotes the two roots of $V_{\infty,M}$, we can check that

$$\begin{aligned} \lim_{R \rightarrow \infty} -V'_{R,M,-1}(r_+) &\approx \frac{n}{(n+1)(M+1)} (e^{s_+} - e^{-ns_+}) \\ &< \frac{n}{(n+1)(M+1)} (e^{-ns_-} - e^{s_-}) \approx \lim_{R \rightarrow \infty} V'_{R,M,-1}(r_-). \end{aligned}$$

By denoting $U_{R,M,-1,f_{-1}} = u_R$ and $r_{\pm}(R, M, -1, f_{-1}) = r_{\pm}(R)$ with $M = \cosh(3) - 1$, in Figure 2 we obtain numerically the sizes of the interval $\text{Gap}_{M,-1,f_{-1}}$ for different values of n .

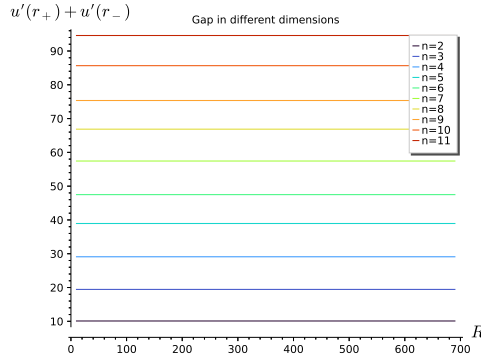


Figure 2: In this graph we picture the function $s(R) = |u'_R(r_-(R)) + u'_R(r_+(R))|$, where u_R is a solution for the Serrin's problem as stated in Remark 2.9 with Cauchy data (2.4), for different dimensions n , which are represented with different colors. Note that these functions are so close to be constants for R big that they look as straight lines in the scale presented in this picture. Observe also that $\mp u'_R(r_{\pm}(R)) = c(M, -1, f_{-1}) \cdot \tau_{M,-1,f_{-1}}^{\pm}(R)$, so each graph have the horizontal line of height $c(M, -1, f_{-1}) \cdot \text{Length}(\text{Gap}_{M,-1,f_{-1}})$ as asymptote.

2.3 Construction of the comparison framework

Let $(\mathcal{M}, g)$ be an n -dimensional complete Riemannian manifold with a Ricci lower bound $\text{Ric}_g \geq (n-1)kg$, and let $\Omega \subset \mathcal{M}$ be a bounded $\mathcal{C}^2$ -domain. Fix $f \in \text{Lip}_{\text{loc}}(\mathbb{R})$ that is k -admissible in the sense of Definition 2.3, and consider a solution (Ω, u) of (1.2).

The problem (1.2) is invariant under isometries: if $\mathcal{I} \in \text{Iso}(\mathcal{M})$, then $(\mathcal{I}(\Omega), u \circ \mathcal{I}^{-1})$ also solves (1.2). We say that (Ω, u) and $(\tilde{\Omega}, \tilde{u})$ are *congruent*, denoted $(\Omega, u) \equiv (\tilde{\Omega}, \tilde{u})$, if there exists $\Phi \in \text{Iso}(\mathcal{M})$ such that $\Phi(\Omega) = \tilde{\Omega}$ and $\tilde{u} = u \circ \Phi^{-1}$ in $\tilde{\Omega}$.

Step 1. Choosing a connected component. Working up to congruence, fix (Ω, u) and set $M := u_{\max} \in \mathcal{I}_f$. Let $\mathcal{U}$ be a connected component of $\Omega \setminus \text{Max}(u)$. Since u is a solution to (1.2) and f is k -admissible, it follows from the maximum principle that $\partial\Omega \cap \text{cl}(\mathcal{U}) \neq \emptyset$.

Definition 2.10. For each connected component $\Gamma \in \pi_0(\partial\Omega)$, define

$$\bar{\tau}(\Gamma) := \max_{p \in \Gamma} \frac{|\nabla u(p)|^2}{c(M, k, f)}.$$

Then, for each $\mathcal{U} \in \pi_0(\Omega \setminus \text{Max}(u))$, set

$$\bar{\tau}(\mathcal{U}) := \max\{\bar{\tau}(\Gamma) : \Gamma \in \pi_0(\partial\Omega \cap \text{cl}(\mathcal{U}))\}.$$

For the whole domain we define,

$$\bar{\tau}(\Omega) := \max\{\bar{\tau}(\mathcal{U}) : \mathcal{U} \in \pi_0(\Omega \setminus \text{Max}(u))\}.$$

Step 2. Construction of the model manifold. Choose a boundary component $\Gamma_0 \in \pi_0(\partial\Omega \cap \text{cl}(\mathcal{U}))$ such that $\bar{\tau}(\Gamma_0) = \bar{\tau}(\mathcal{U})$. We then work on the model manifold whose fiber is Γ_0 equipped with the metric induced by g on Γ_0 , denoted by $\bar{g}$:

$$(\mathcal{M}_k^n(\Gamma_0), g_k) := (I_k \times \Gamma_0, dr^2 + s_k(r)^2 \bar{g}).$$

This choice fixes the reference geometry for the comparison on $\mathcal{U}$.

Step 3. Comparison criteria within the model family. Within the model pairs on $\mathcal{M}_k^n(\Gamma_0)$ with maximum value M , we single out those whose normalized boundary response matches (or dominates) $\bar{\tau}(\mathcal{U})$.

Definition 2.11 (Comparison pair and associated model pair). We say that a model pair

$$(\bar{\mathcal{U}}, \bar{u}) := (\mathcal{U}_{R,M,k,f}^\diamond, u_{R,M,k,f}^\diamond) \in \text{Model}_{M,k,f}^\diamond, \quad \diamond \in \{-, +\},$$

in $\mathcal{M}_k^n(\Gamma_0)$ is a comparison pair for $(\mathcal{U}, u)$ if

$$\bar{\tau}(\mathcal{U}) \leq \bar{\tau}(\bar{\mathcal{U}}) = \bar{\tau}_{M,k,f}^\diamond(R), \quad (2.11)$$

and an associated model pair if equality holds. In either case, we call $\bar{u}$ a comparison function for u in $\mathcal{U}$.

In Subsection 2.3.1 we discuss existence and non-uniqueness of comparison pairs and associated model pairs, and we record consequences of the Standard Conditions.

Step 4. Construction of the comparison tool. Fix a comparison pair $(\bar{\mathcal{U}}, \bar{u})$ for $(\mathcal{U}, u)$. Denote by $\bar{U}$ the radial profile generating $\bar{u}$, by $\bar{R}$ its core radius, and write $\bar{r}_\pm := r_\pm(\bar{R}, M, k, f)$. Following [?, Subsection 5.4], define $G : [0, M] \times [\bar{r}_-, \bar{r}_+] \rightarrow \mathbb{R}$ by

$$G(s, r) := s - \bar{U}(r).$$

Then $\partial_r G(s, r) = 0$ if and only if $\bar{U}'(r) = 0$, i.e. $r = \bar{R}$ (recall Definition 2.3). By the Implicit Function Theorem, there exist two $\mathcal{C}^2$ -functions

$$\chi^- : [0, M) \rightarrow [\bar{r}_-, \bar{R}], \quad \chi^+ : [0, M) \rightarrow [\bar{R}, \bar{r}_+], \quad (2.12)$$

such that $G(s, \chi^\pm(s)) = 0$ for all $s \in [0, M)$, i.e. $\bar{U}(\chi^\pm(s)) = s$. Equivalently, $\chi^\pm$ is the inverse of $\bar{U}$ on the corresponding monotonicity interval. We use these inverses to define a pseudo-radial map on $\mathcal{U}$, which will allow us to compare the geometry of level sets in Section 3.

Definition 2.12 (Pseudo-radial function). *Let $(\bar{\mathcal{U}}, \bar{u})$ be a comparison pair for $(\mathcal{U}, u)$ with sign $\diamond \in \{-, +\}$. The pseudo-radial function associated to $(\mathcal{U}, u)$ and $(\bar{\mathcal{U}}, \bar{u})$ is*

$$\Psi : \mathcal{U} \rightarrow \chi^\diamond(0, M) \subset [\bar{r}_-, \bar{r}_+], \quad \Psi(p) := \chi^\diamond(u(p)).$$

Remark 2.13. *From the definition of the function $\chi^\diamond$, it is clear that it extends continuously to $[0, M]$. Furthermore, since $u \in \mathcal{C}^2(\Omega)$, the function Ψ is $\mathcal{C}^2$ on $\mathcal{U}$ away from the critical set of u , and Ψ is Lipschitz on compact subsets of $\mathcal{U}$. Moreover, by construction $\bar{U} \circ \Psi = u$ on $\mathcal{U}$.*

At this point the role of k -admissibility becomes clear: it ensures that for each prescribed maximum value M and core radius R , the corresponding radial Cauchy solution is positive between its first and last zeros, strictly monotone away from the maximum set, and vanishes at the boundary radii. These properties allow us to invert the profile on each branch and to define the pseudo-radial map Ψ for a general solution.

2.3.1 Existence and non-uniqueness of associated model pairs

We record a few remarks about comparison/associated model pairs and about the admissible range of the parameter $\bar{\tau}$.

On non-uniqueness. An associated model pair in Definition 2.11 need not be unique: if the map $R \mapsto \bar{\tau}_{M,k,f}^\diamond(R)$ is not injective, distinct core radii can produce the same $\bar{\tau}$ -value. This does not affect the arguments below, which only require the existence of at least one comparison (or associated) pair.

Existence issues. By definition, an associated model pair exists if and only if $\bar{\tau}(\mathcal{U}) \in \text{Adm}_{M,k,f}$. For a general k -admissible f we do not know whether this always holds. Nevertheless, comparison pairs exist under mild lower bounds on f , and associated model pairs exist in many cases of interest; see Lemma 2.16 below.

As a consequence of Proposition 2.7, we obtain the following existence statement.

Corollary 2.14. *Let $f \in \text{Lip}_{\text{loc}}(\mathbb{R})$ be a function that satisfies the Standard Conditions. Suppose that $\Omega \Subset \mathcal{M}$ is a bounded domain and that u solves (1.2). Then for every $\mathcal{U} \in \pi_0(\Omega \setminus \text{Max}(u))$, there exists at least one comparison pair for $(\mathcal{U}, u)$. In the particular case of $k > 0$, for every $\bar{\tau}(\mathcal{U}) > \tau_{k,f}^0(M)$ there is an associated model pair.*

We will also use the following notion of regular point.

Definition 2.15. *Let $A \subset \mathcal{M}$ and let $l \in \{1, \dots, n\}$. We say that $p \in A$ is an l -dimensional regular point if A is an l -dimensional C^1 -submanifold in a neighbourhood of p .*

Under assumptions stronger than the Standard Conditions, the comparison method becomes effective in the sense that regularity information on $\text{Max}(u) \cap \text{cl}(\mathcal{U})$ forces $\bar{\tau}(\mathcal{U})$ to lie in the admissible range. This generalizes [?, Proposition 4.3] (which in turn is a version of [?, Theorem 3.3]).

Lemma 2.16. *Let $f \in C^m(\mathbb{R})$ with $m := \max\{n - 2, 3\}$ be a k -admissible function with $f(0) > 0$ if $k \leq 0$, and assume*

$$f'(x) \geq nk \quad \text{for all } x \in \mathcal{I}_f. \quad (2.13)$$

Then:

1. *If $\text{Max}(u) \cap \text{cl}(\mathcal{U})$ contains an $(n - 1)$ -dimensional regular point, then $\bar{\tau}(\mathcal{U}) > \bar{\tau}_{k,f}^0(M)$.*
2. *If $(\mathcal{M}, g) = \mathbb{M}^n(k)$, then $\bar{\tau}(\mathcal{U}) \notin \text{Gap}_{M,k,f}$.*

In particular, if $(\mathcal{M}, g) = \mathbb{M}^n(k)$ and $\text{Max}(u) \cap \text{cl}(\mathcal{U})$ contains a smooth hypersurface, then there exists an associated model pair for $(\mathcal{U}, u)$ with positive core radius.

We postpone the proof of this result to Section 3.1, where (2.13) will also be crucial for obtaining the gradient estimate away from $\text{Max}(u)$.

To clarify how the hypothesis packages on f feed into the properties of the $\bar{\tau}$ -functions and the existence of comparison/associated model pairs, Figure 3 summarizes the logical dependencies.

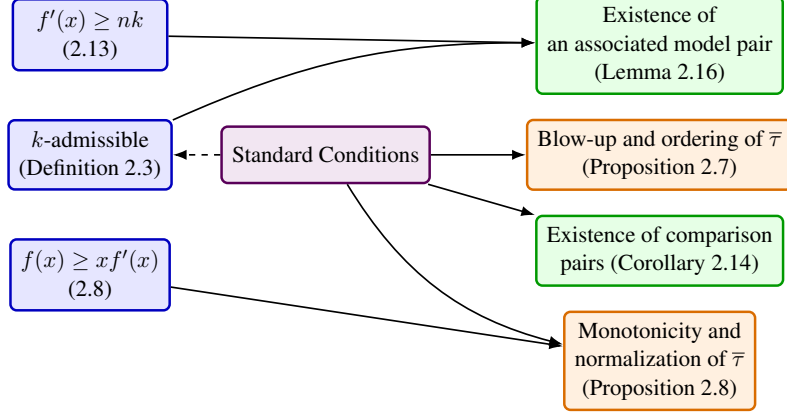


Figure 3: The diagram summarizes the dependencies between the results previously stated and the conditions over the non-linearity. Arrows indicate requirements (the arrows point to the conditions necessary to obtain that result); dashed arrows indicate sufficient conditions. Orange boxes correspond to properties of the $\bar{\tau}$ -functions, green boxes to existence statements for comparison/model pairs, and purple/blue boxes to hypothesis on f .

3 Proof of the Comparison Results

Let $(\mathcal{M}^n, g)$ be a complete Riemannian manifold with the lower Ricci bound (1.1), and let $f \in \text{Lip}_{loc}(\mathbb{R})$ be k -admissible. Fix a solution (Ω, u) of (1.2) and choose a connected component

$$\mathcal{U} \in \pi_0(\Omega \setminus \text{Max}(u)) \quad \text{with} \quad \bar{\tau}(\mathcal{U}) = \bar{\tau}(\Omega).$$

In this section we prove the gradient comparison (Theorem A) and then derive the geometric estimates stated in Section 2. Technical computations are deferred to Appendix B.

Proof of Theorem A. In order to simplify the notation we write

$$W = |\nabla u|^2 \quad \text{and} \quad \bar{W} = |\nabla \bar{u}|^2 \circ \Psi \quad \text{in } \mathcal{U}.$$

By Lemma B.1 (Appendix B), the function

$$F_\beta = \left(\frac{s_k(\Psi)}{\bar{U}'(\Psi)} \right)^{2\frac{n-1}{n}} (W - \bar{W})$$

where s_k is given by (2.1) and F_β satisfies the elliptic inequality

$$\Delta F_\beta - 2\frac{n-1}{n}\lambda(\Psi) \langle \nabla F_\beta, \nabla u \rangle - 2\frac{n-1}{n} \frac{|\nabla u|^2}{\bar{U}'(\Psi)^2} \mu(\Psi) F_\beta \geq 0,$$

with μ being non-negative in $\mathcal{U}$. Moreover, F_β extends continuously to $\text{cl}(\mathcal{U})$ and satisfies $F_\beta \leq 0$ on $\partial\mathcal{U}$. Indeed, on $\partial\Omega \cap \text{cl}(\mathcal{U})$ we have $u = 0$, hence Ψ is constant and $\bar{W}$ equals the corresponding model boundary gradient; the comparison-pair condition (2.11) implies $W \leq \bar{W}$ there, and therefore $F_\beta \leq 0$ on $\partial\Omega \cap \text{cl}(\mathcal{U})$. On the remaining portion $\text{cl}(\mathcal{U}) \cap \text{Max}(u)$, one has $W = 0$ and $\bar{W} = 0$; in addition, the weight in the definition of F_β is chosen precisely so that F_β has a finite continuous trace and vanishes on this set (see the proof of Lemma B.1).

Applying the maximum principle, we get that $W \leq \bar{W}$ on $\mathcal{U}$. If equality holds at one point, then the strong maximum principle forces $F_\beta \equiv 0$, and therefore $W = \bar{W}$ on $\mathcal{U}$. Inspecting the inequality in Lemma B.1 then yields

$$\text{Ric}(\nabla u, \nabla u) = (n-1)k|\nabla u|^2 \text{ on } \mathcal{U}.$$

Moreover, by the proof of Lemma B.1 one also obtains the identity at each level set:

$$\nabla^2 u = -\lambda(\Psi) du \otimes du + \frac{1}{n} \left(\lambda(\Psi) |\nabla u|^2 - f(u) \right) g_0,$$

where λ is given by (B.3) and g_0 is the induced metric on Γ_0 (recall Definition 2.11). Hence, Brinkmann's Theorem (see [?, Theorem 4.3.3]) shows that the metric on $\mathcal{U}$ must then split as

$$g = d\Psi^2 + s_k(\Psi)^2 \bar{g} \quad \text{on } \mathcal{U},$$

so $(\mathcal{U}, u)$ is congruent to the model pair $(\bar{\mathcal{U}}, \bar{u})$, as claimed. $\square$

Remark 3.1. If $\bar{\tau}(\mathcal{U}) = \bar{\tau}(\bar{\mathcal{U}}) = 1$, extending Ψ to $\text{Max}(u) \cap \text{cl}(\mathcal{U})$, we obtain

$$\nabla^2 S_k(\Psi) = s'_k(\Psi) g,$$

with $S_k \circ \Psi = 1$ and $\nabla(S_k \circ \Psi) = 0$ on this set. It then follows again from [?, Theorem 4.3.3] that $g_0 = g_{\mathbb{S}^{n-1}}$ and $\Gamma = \mathbb{S}^n$. This, in turn, implies that

$$\text{Max}(u) = \{o\}, \quad (\mathcal{M}, g) = \mathbb{M}^n(k),$$

Ω is a ball, and u is radially symmetric with respect to o .

About the conditions of Theorem A

As mentioned in Subsection 1.2, we do not claim that the hypotheses in Theorem A are optimal. For instance, the torsional rigidity problem on $\mathbb{S}^2$ treated in [?] does not satisfy these assumptions, yet gradient estimates can still be obtained in that setting.

On $\mathbb{S}^3$ with the round metric, one can also write explicit radial solutions for affine Helmholtz nonlinearities $f(x) = \lambda x + \beta$, with $\lambda \in (-1, 0) \cup (0, \pi)$ (excluding singular cases) and $\beta \in \mathbb{R}^+$ chosen so that f is k -admissible:

$$u(r) = \frac{\csc(r) \sin(R)(\lambda M + \beta) \cos(\sqrt{\lambda+1}(r-R)) + \frac{\csc(r) \cos(R)(\lambda M + \beta) \sin(\sqrt{\lambda+1}(r-R))}{\sqrt{\lambda+1}} - \beta}{\lambda}.$$

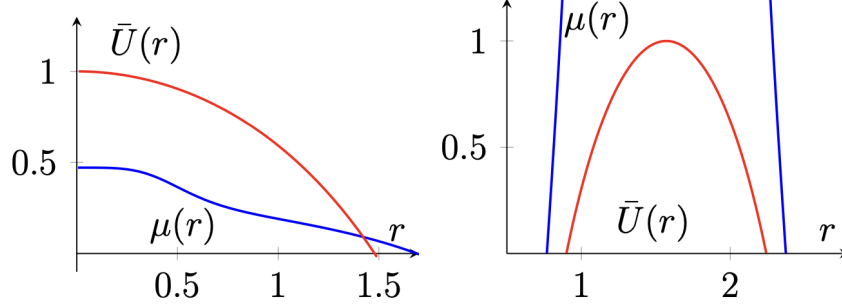


Figure 4: Comparison of the radial profile $\bar{U}$ (red) and the function $\mu(r)$ defined in (B.4) (blue) for affine Helmholtz equations on the three-dimensional sphere that do not satisfy condition (2.8). *Left:* Let $f(x) = \frac{-1}{4}x + \frac{5}{2}$ and consider the radial solution with maximum 1. For this f we compute numerically that $\mu(r)$ is positive independently of the initial conditions. *Right:* Let $f(x) = x + 2.9$ and consider the solution with maximum 1 at radius $\frac{\pi}{2}$. For this affine term we verified (with Sage) that $\mu(r)$ is positive only for certain initial conditions.

For these explicit solutions, we observe that for certain initial data and suitable choices of (λ, β) the quantity $\mu(r)$ in Lemma B.1 is positive, as in the right panel of Figure 4. Moreover, there are parameter ranges (λ, β) for which μ in Lemma B.1 remains positive independently of the initial data, as illustrated in the left panel of Figure 4. However, in other cases, such as Allen–Cahn-type nonlinearities, one finds that if $\bar{U}(r)$ is a solution on $\mathbb{S}^n$ for $n \geq 2$, then at the boundary radii $\bar{r}_\pm$ one has $\mu(\bar{r}_\pm) = 1 - nk < 0$, independently of the initial conditions.

3.1 Curvature estimates

In this subsection we extend the curvature estimates in [?, Subsection 4.2] to higher dimensions and prove Lemma 2.16. Throughout, we work under the assumptions of Theorem A.

Proposition 3.2. *Let f be a k -admissible function satisfying the conditions (2.8) and (2.13). Let $(\bar{\mathcal{U}}, \bar{u})$ be an associated model pair to $(\mathcal{U}, u)$ in the sense of Definition 2.11, and consider $p \in \partial\Omega$ such that*

$$|\nabla u|^2(p) = \max_{x \in \partial\Omega \cap \text{cl}(\mathcal{U})} |\nabla u|^2(x). \quad (3.1)$$

Let $\bar{R} \in \mathcal{R}_f$ be the core radius of $(\bar{\mathcal{U}}, \bar{u})$. Then, it holds

$$H(p) \leq \cot_k(\bar{r}_+) \quad \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+ \quad \text{and} \quad H(p) \leq -\cot_k(\bar{r}_-) \quad \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^-,$$

where $\bar{r}_-$ and $\bar{r}_+$ are defined as the zeros of the solution to (2.3)-(2.4) defining $(\bar{\mathcal{U}}, \bar{u})$ and $H(p)$ is computed with respect to the inner orientation to $\mathcal{U}$.

Proof. Let $p \in \partial\Omega \cap \text{cl}(\mathcal{U})$ satisfy (3.1), and let $\gamma : [0, \varepsilon) \rightarrow \mathcal{M}$ be the unit-speed geodesic with $\gamma(0) = p$ and $\gamma'(0) = \nabla u / |\nabla u|(p)$. As in [?, Proposition 5.1], for t small one has

$$W(\gamma(t)) = W(p) + 2\left((n-1)\sqrt{W}(p)H(p) - f(0)\right)\sqrt{W}(p)t + O(t^2),$$

and

$$\bar{W}(\gamma(t)) = \bar{W}(p) + 2\left(\mp(n-1)\cot_k(\bar{r}_\pm)\sqrt{\bar{W}}(p) - f(0)\right)\sqrt{\bar{W}}(p)t + O(t^2),$$

where we take $\bar{r}_+$ when $(\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+$ and $\bar{r}_-$ otherwise. Since (3.1) implies $W(p) = \bar{W}(p)$, the claim follows by comparing the first-order terms and using Theorem A. $\square$

Remark 3.3. *The conclusion of Proposition 3.2 requires an associated model pair: only in this case do we have $W(p) = \bar{W}(p)$ at a boundary point satisfying (3.1).*

Proposition 3.4. *Let $f \in \mathcal{C}^m(\mathbb{R})$ with $m := \max\{n-2, 3\}$ be a k -admissible function satisfying (2.13). Let $(\bar{\mathcal{U}}, \bar{u})$ be a comparison pair with core radius $\bar{R} > 0$, and let $p \in \text{cl}(\mathcal{U}) \cap \text{Max}(u)$ be a $(n-1)$ -dimensional regular point. Then $\text{Max}(u)$ is a $\mathcal{C}^{m+2}$ -hypersurface in a neighbourhood of p , and*

$$H(p) \leq -\cot_k(\bar{R}) \quad \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+ \quad \text{and} \quad H(p) \leq \cot_k(\bar{R}) \quad \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^-,$$

where $H(p)$ is computed with respect to the inner orientation of $\mathcal{U}$.

Proof. Since $u = 0$ along $\partial\Omega$ and $u > 0$ in Ω , we have $p \in \Omega$. Since p is a regular point, there exists a neighbourhood $\mathcal{V} \subset \Omega$ of p such that $\Sigma := \mathcal{V} \cap \text{Max}(u)$ is an embedded, two-sided $\mathcal{C}^1$ hypersurface dividing $\mathcal{V} \setminus \Sigma$ into two connected components $\mathcal{V}_+$ and $\mathcal{V}_-$, with $\mathcal{V}_+ \subset \mathcal{U}$. Since $f \in \mathcal{C}^m$, elliptic regularity gives $u \in \mathcal{C}^{m+2}$ in Ω , and [?, Theorem 2] implies that Σ is $\mathcal{C}^{m+2}$ in a (possibly smaller) neighbourhood of p .

Denote again this neighbourhood by $\mathcal{V}$ and define the signed distance function

$$s(x) := \begin{cases} +\text{dist}(x, \Sigma) & \text{if } x \in \mathcal{V}_+, \\ -\text{dist}(x, \Sigma) & \text{if } x \in \mathcal{V}_-. \end{cases}$$

Since u is at least $\mathcal{C}^5$ in Ω , we can use the fourth-order expansion in [?, Theorem 3.1] and reproduce the argument of [?, Proposition 5.2] to obtain, around $p \in \Sigma$,

$$W = f(M)^2 s^2 (1 + (n-1)H(p)s) + O(s^4), \quad (3.2)$$

$$\bar{W} = f(M)^2 s^2 \left(1 + (n-1) \left(\frac{H(p)}{3} \pm \frac{2\cot_k(\bar{R})}{3} \right) s \right) + O(s^4), \quad (3.3)$$

where $H(p)$ is computed with respect to the normal pointing into $\mathcal{V}_+$, and in (3.3) we take the positive sign when $(\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+$ and the negative sign otherwise. The claim then follows as in Proposition 3.2, using that $W \leq \bar{W}$ in $\mathcal{V}_+$ by Theorem A. $\square$

As a consequence of Proposition 3.4, we can now prove Lemma 2.16.

Proof of Lemma 2.16. We first prove the *item 1*. The argument is the same as in [?, Proposition 5.3], but we include it for completeness. Assume that there exists a $(n-1)$ -dimensional regular point $p \in \text{Max}(u) \cap \bar{\mathcal{U}}$ such that u is $\mathcal{C}^{m+2}$, $m+2 \geq 5$, near p . Arguing by contradiction, suppose that $\bar{\tau}(\mathcal{U}) \leq \bar{\tau}_{M,k,f}(0)$. Then Proposition 3.4 implies that $\text{Max}(u)$ is $\mathcal{C}^{m+2}$ near p and that its mean curvature satisfies

$$H(p) \leq -\cot_k(R) \text{ for all } R \in (0, \bar{r}_k),$$

since every pair in $\text{Model}_{M,k,f}^+$ is a comparison pair. Letting $R \rightarrow 0^+$ yields that $H(p)$ is unbounded, a contradiction. Therefore, $\bar{\tau}(\mathcal{U}) > \tau_{k,f}^0(M) = \inf_{R \in [0, \bar{r}_k)} \bar{\tau}_{M,k,f}^+(R)$.

We now prove the *item 2*. Let $(\mathcal{M}, g) = \mathbb{M}^n(k)$ with $k \leq 0$. We show that

$$\bar{\tau}(\mathcal{U}) \notin \text{Gap}_{M,k,f}.$$

Again, we argue by contradiction.

Since $f \in \mathcal{C}^m$ where $m := \max\{n-2, 3\}$ satisfies the Standard Conditions and $M \in \mathcal{I}_f$, the maximum principle ensures that $\text{cl}(\mathcal{U}) \cap \partial\Omega \neq \emptyset$. Choose a connected component $\Gamma \subset \text{cl}(\mathcal{U}) \cap \partial\Omega$. Because Γ is part of a level set of a $\mathcal{C}^{m+2}$ function and Ω is bounded, Γ is a properly embedded, compact $\mathcal{C}^{m+2}$ hypersurface in $\mathbb{M}^n(k)$. By the Jordan–Brouwer Separation Theorem, $\mathbb{M}^n(k) \setminus \Gamma = \mathcal{V}_1 \cup \mathcal{V}_2$, where $\mathcal{V}_1$ is open and bounded. Thus either $\mathcal{U} \subset \mathcal{V}_1$ or $\mathcal{U} \subset \mathcal{V}_2$.

Case 1: $\mathcal{U} \subset \mathcal{V}_1$. For any point $p \in \mathcal{U}$, let $u_{R,M,k,f}$ denote the model solution centered at p , written in terms of $r_p = \text{dist}(p, \cdot)$. We use the notation $\mathcal{U}_{R,M,k,f}^\pm$ from Section 2.1 for the regions where $u_{R,M,k,f}$ is positive and does not attain its maximum. Choose $R_0 > 0$ so large that $\mathcal{U}_{R_0,M,k,f}^+ \cap \mathcal{V}_1 = \emptyset$. For $R < R_0$, set

$$w = u_{R,M,k,f} - u \quad \text{in} \quad \mathcal{U}_{R,M,k,f}^+ \cap \mathcal{U}.$$

By decreasing R slightly from R_0 , we reach a first-contact configuration such that

1. $w \leq 0$ in $\mathcal{U}_{R,M,k,f}^+ \cap \mathcal{U}$,
2. there exists a point q either in $\mathcal{U}_{R,M,k,f}^+ \cap \mathcal{U}$ or in $\Gamma_{R,M,k,f}^+ \cap \partial\mathcal{U}$ such that $w(q) = 0$.

Since $\bar{\tau}(\mathcal{U}_{R,M,k,f}^+) < \bar{\tau}(\mathcal{U})$, the strong maximum principle implies that $u_{R,M,k,f} \equiv u$ in a neighbourhood of q , a contradiction.

Case 2: $\mathcal{U} \subset \mathcal{V}_2$. Choose $p \in \mathcal{V}_1$ and $R_0 > 0$ so that $\mathcal{U}_{R_0,M,k,f}^- \subset \mathcal{V}_1$. Define

$$w = u - u_{R,M,k,f} \quad \text{in} \quad \mathcal{U}_{R,M,k,f}^- \cap \mathcal{U},$$

and take $R > R_0$ large. The same first-contact argument yields a contradiction with the strong maximum principle.

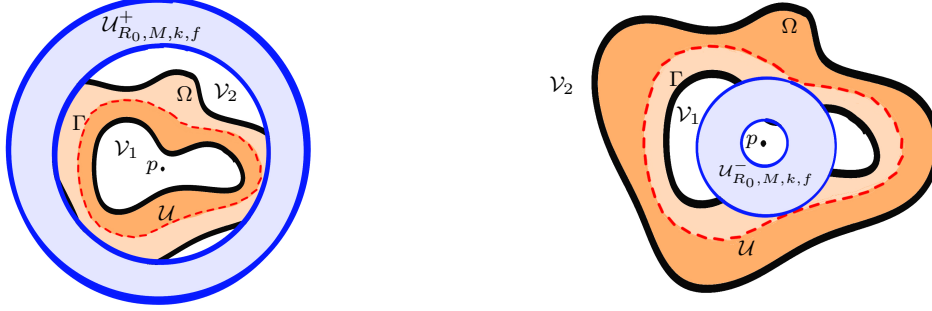


Figure 5: Schematic representation of the two comparison configurations used in the proof of Lemma 2.16. The dark orange region represents $\mathcal{U}$, the dashed red hypersurface represents Γ , and the blue region denotes the comparison region centered at p . Left: Case 1, where $\mathcal{U} \subset \mathcal{V}_1$ and the family $\mathcal{U}_{R_0,M,k,f}^+$ is shrunk from outside $\mathcal{V}_1$. Right: Case 2, where $\mathcal{U} \subset \mathcal{V}_2$ and the family $\mathcal{U}_{R_0,M,k,f}^-$ is expanded from $\mathcal{V}_1$.

Both cases are impossible, hence

$$\bar{\tau}(\mathcal{U}) \notin \text{Gap}_{M,k,f}.$$

□

3.2 An isoperimetric inequality

In this subsection we prove Theorem B for a component $(\mathcal{U}, u)$ such that $\text{Max}(u) \cap \text{cl}(\mathcal{U})$ contains a Lipschitz-continuous hypersurface. The key input is an area bound for the top level set, extending [?, Proposition 5.4] to arbitrary dimension and to general level sets.

Throughout, we keep the standing assumptions on $(\mathcal{M}, g)$ and assume that $f \in \mathcal{C}^m(\mathbb{R})$, where $m = \max\{n - 2, 3\}$, is k -admissible and satisfies (2.13).

Proposition 3.5. *Let $(\bar{\mathcal{U}}, \bar{u})$ be a comparison pair with core radius $\bar{R} > 0$. Let $t \in [0, M)$ be a regular value of u and set $\Gamma_t := u^{-1}(t)$. Assume that the $(n - 1)$ -dimensional part of $\text{cl}(\mathcal{U}) \cap \text{Max}(u)$, denoted by $\Gamma_M \subset \text{cl}(\mathcal{U}) \cap \text{Max}(u)$, is a Lipschitz-continuous hypersurface.*

Then,

$$\mathcal{H}^{n-1}(\Gamma_M) \leq \begin{cases} \left(\frac{s_k(\bar{R})}{s_k(\chi_+(t))} \right)^{n-1} \mathcal{H}^{n-1}(\Gamma_t) & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+, \\ \left(\frac{s_k(\bar{R})}{s_k(\chi_-(t))} \right)^{n-1} \mathcal{H}^{n-1}(\Gamma_t) & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^-, \end{cases} \quad (3.4)$$

where $\chi_{\pm}$ is given in (2.12). Furthermore, equality holds in (3.4) for some t if, and only if $(\mathcal{U}, u) \equiv (\bar{\mathcal{U}}, \bar{u})$.

Proof. The argument follows [?, Proposition 2.4], which treats the case $t = 0$ for Serrin's equation in $\mathbb{R}^n$. Define the vector field

$$\mathcal{X}(p) := \frac{1}{s_k(\Psi)^n \bar{U}'(\Psi)} \nabla u \quad \text{for all } p \in \Omega \setminus \text{Max}(u), \quad (3.5)$$

whose divergence is

$$\text{div}(\mathcal{X}) = \frac{1}{s_k(\Psi)^n \bar{U}'(\Psi)^3} (f(u)(W - \bar{W}) - \cot_k(\Psi)W).$$

Since f is of class $\mathcal{C}^m$, it follows that $u \in \mathcal{C}^{m+2}(\mathcal{U})$, so Sard's Theorem implies that the set of critical values of u in $\mathcal{U}$ has measure zero. Hence, we can fix $\varepsilon > 0$ arbitrarily small such that $M - \varepsilon$ is a regular value of u , and set

$$\mathcal{U}_{t,\varepsilon} = \{p \in \mathcal{U} : t < u(p) < M - \varepsilon\}.$$

Applying the divergence theorem in $\mathcal{U}_{t,\varepsilon}$ gives

$$\begin{aligned} \int_{\mathcal{U}_{t,\varepsilon}} \frac{1}{s_k(\Psi)^n \bar{U}'(\Psi)^3} \left(f(u)(W - \bar{W}) - \cot_k(\Psi)W \right) d\mu = \\ \int_{\Gamma_t} \frac{-|\nabla u|}{s_k(\Psi)^n \bar{U}'(\Psi)} d\sigma + \int_{\Gamma_{M-\varepsilon}} \frac{|\nabla u|}{s_k(\Psi)^n \bar{U}'(\Psi)} d\sigma, \end{aligned} \quad (3.6)$$

where $d\mu$ and $d\sigma$ are the volume elements of $\mathcal{M}$ and the induced hypersurface measure, respectively, and the inner unit normal satisfies $\nu = \nabla u / |\nabla u|$ along Γ_t and $\nu = -\nabla u / |\nabla u|$ along $\Gamma_{M-\varepsilon}$.

Along Γ_t the pseudo-radial function is constant, $\Psi = \chi_{\pm}(t)$, and Theorem A gives $W \leq \bar{W}$. Therefore,

$$\int_{\Gamma_t} \frac{|\nabla u|}{s_k(\Psi)^n \bar{U}'(\Psi)} d\sigma \begin{cases} \geq -\frac{\mathcal{H}^{n-1}(\Gamma_t)}{s_k^{n-1}(\chi_+(t))} & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+, \\ \leq \frac{\mathcal{H}^{n-1}(\Gamma_t)}{s_k^{n-1}(\chi_-(t))} & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^-. \end{cases} \quad (3.7)$$

Since Γ_M is Lipschitz, it is $(n-1)$ -rectifiable and has differentiability points of full $\mathcal{H}^{n-1}$ -measure. Using [?, Theorem 2] and the expansions (3.2)–(3.3), we have

$$\lim_{x \in \mathcal{U}, x \rightarrow p} \frac{W}{\bar{W}} = 1 \quad \text{for } \mathcal{H}^{n-1}\text{-a.e. } p \in \Gamma_M.$$

Taking a sequence $\varepsilon_n \rightarrow 0$ such that $M - \varepsilon_n$ is a regular value for all n , we obtain

$$\lim_{n \rightarrow +\infty} \int_{\Gamma_{M-\varepsilon_n}} \frac{|\nabla u|}{s_k(\Psi)^n \bar{U}'(\Psi)} d\sigma = \begin{cases} -\frac{\mathcal{H}^{n-1}(\Gamma_M)}{s_k^{n-1}(\bar{R})} & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+, \\ \frac{\mathcal{H}^{n-1}(\Gamma_M)}{s_k^{n-1}(\bar{R})} & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^-. \end{cases} \quad (3.8)$$

Finally, since $\bar{U}'(\Psi)^3 < 0$ on $\mathcal{U}$ if $(\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+$ and $\bar{U}'(\Psi)^3 > 0$ if $(\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^-$, we conclude that

$$\int_{\mathcal{U}} \frac{1}{s_k(\Psi)^n \bar{U}'(\Psi)^3} \left(f(u)(W - \bar{W}) - \cot_k(\Psi)W \right) d\mu \begin{cases} \geq 0 & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+, \\ \leq 0 & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^-. \end{cases} \quad (3.9)$$

Letting $\varepsilon_n \rightarrow 0$ in (3.6) and using (3.7)–(3.9) yields (3.4).

For rigidity, note that equality in (3.4) forces equality in (3.9), hence $W = \bar{W}$ on $\mathcal{U}$, and then $(\mathcal{U}, u) \equiv (\bar{\mathcal{U}}, \bar{u})$ by the rigidity statement in Theorem A. $\square$

Combining Proposition 3.5 with the coarea formula yields Theorem B.

Proof of Theorem B. We only consider the case $(\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+$. The case $(\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^-$ is identical.

By the Unique Continuation principle for u , the set of critical points cannot have interior points; thus the hypotheses of [?, Proposition 2.2] are satisfied, and the coarea formula can be applied yielding,

$$\mathcal{H}^n(\mathcal{U}) = \int_{(0,M)} \left(\int_{\Gamma_t} \frac{1}{|\nabla u|} d\sigma_t \right) dt,$$

where $d\sigma_t$ denotes the induced area element on Γ_t . Using Theorem A and (??), we obtain

$$\mathcal{H}^n(\mathcal{U}) \geq \int_0^M \left(\int_{\Gamma_t} \frac{1}{\sqrt{\bar{W}}} d\sigma_t \right) dt, \quad (3.10)$$

since $\bar{W} > 0$ on Γ_t for $t < M$.

Moreover, $\bar{W}$ is constant along each Γ_t and

$$\bar{W} = \bar{U}'(\chi_{\pm}(t))^2, \quad t \in (0, M),$$

where we take χ_+ when $\bar{\tau}(\mathcal{U}) \leq \tau_k^+(M)$ and χ_- otherwise. Therefore,

$$\begin{aligned} \int_0^M \left(\int_{\Gamma_t} \frac{1}{\sqrt{W}} d\sigma_t \right) dt &= \int_0^M \frac{\mathcal{H}^{n-1}(\Gamma_t)}{|\bar{U}'(\chi_+(t))|} dt \\ &\geq \frac{\mathcal{H}^{n-1}(\Gamma_M)}{s_k(\bar{R})^{n-1}} \int_0^M \frac{s_k(\chi_+(t))^{n-1}}{|\bar{U}'(\chi_+(t))|} dt, \end{aligned} \quad (3.11)$$

where we used Proposition 3.5 in the last inequality.

Finally, since $\chi_+ : [0, M] \rightarrow [\bar{R}, \bar{r}_+]$ is decreasing, the change of variables $t = \chi_+^{-1}(r)$ yields

$$\int_0^M \frac{s_k(\chi_+(t))^{n-1}}{|\bar{U}'(\chi_+(t))|} dt = \int_{\bar{R}}^{\bar{r}_+} s_k(r)^{n-1} dr.$$

Now observe that

$$\mathcal{H}_k^n((r_1, r_2) \times \Gamma_0) = \mathcal{H}_k^{n-1}(\Gamma_0) \cdot \int_{r_1}^{r_2} s_k(r)^{n-1} dr$$

and

$$\mathcal{H}_k^{n-1}(\{r_1\} \times \Gamma_0) = \mathcal{H}_k^{n-1}(\Gamma_0) \cdot s_k(r_1)^{n-1},$$

for all $r_1, r_2 \in I$ with $r_1 < r_2$.

Hence, combining this with (3.10)–(3.11) gives (1.4). The rigidity statement follows as in Proposition 3.5 $\square$

On the extremal case

We comment on Proposition 3.5 and Theorem B when $\bar{\tau}(\mathcal{U}) \notin \text{Adm}_{M,k,f}$. If $\text{Max}(u) \cap \text{cl}(\mathcal{U})$ contains a Lipschitz hypersurface, then it has differentiability points of full $(n-1)$ -dimensional measure; in particular, it contains an $(n-1)$ -dimensional regular point. Lemma 2.16 then yields

$$\bar{\tau}(\mathcal{U}) > \tau_{k,f}^0(M),$$

so $\bar{\tau}(\mathcal{U}) \notin \text{Adm}_{M,k,f}$ forces $\bar{\tau}(\mathcal{U}) \in \text{Gap}_{M,k,f}$.

When $k > 0$, Proposition 2.7 implies $\text{Gap}_{M,k,f} = \emptyset$, hence the bounds above are always sharp. Assume now $k \leq 0$. By Theorem A, the sharpest estimates are obtained by choosing $\bar{\tau}(\bar{\mathcal{U}}) \in \text{Im}(\bar{\tau}_{M,k,f}^-)$ as close as possible to $\tau_{k,f}^-(M)$. If no $R \in [0, \infty)$ realizes the infimum,

$$\bar{\tau}_{M,k,f}^-(R) = \tau_{k,f}^-(M),$$

then necessarily

$$\lim_{R \rightarrow \infty} \bar{\tau}_{M,k,f}^-(R) = \tau_{k,f}^-(M),$$

which occurs, for instance, under (2.8); see Proposition 2.8. In that case, evaluating (2.3) at $r_-(R, M, k, f)$ gives

$$0 = U_{R,M,k,f}(r_-(R, M, k, f)) = M - \int_{r_-(R,M,k,f)}^R \frac{1}{s_k(y)^{n-1}} \left(\int_R^y s_k(x)^{n-1} f(U_{R,M,k,f}(x)) dx \right) dy,$$

and hence

$$\lim_{R \rightarrow \infty} r_-(R, M, k, f) = +\infty.$$

Letting $R \rightarrow \infty$ in (3.4) and (1.4) then yields, in particular,

$$\mathcal{H}^{n-1}(\Gamma_M) \leq \mathcal{H}^{n-1}(\Gamma_t), \quad \forall t \in [0, M],$$

since

$$\lim_{R \rightarrow \infty} \frac{s_k(R)}{s_k(r_-(R, M, k, f))} = 1.$$

Indeed, if this limit were different from 1, then

$$\lim_{R \rightarrow \infty} \frac{\int_{r_-(R,M,k,f)}^R s_k(r)^{n-1} dr}{s_k(R)^{n-1}} = +\infty,$$

and Theorem B would force $\mathcal{U}$ to be unbounded, a contradiction. A similar argument also excludes the possibility that the ratio tends to 0. Thus Theorem B remains non-trivial in this limiting regime, although the limiting inequality is not explicit in general.

3.3 Location of the hot-spots

In this subsection we estimate the distance from the maximum set of a solution (Ω, u) of (1.2) to the boundary, extending the sharp bounds in [?, Section 5].

Proposition 3.6. *Let $f \in C^1(\mathbb{R})$ be a k -admissible function satisfying (2.8). Let $p \in \text{Max}(u) \cap \text{cl}(\mathcal{U})$. Then*

$$\text{dist}(p, \partial\Omega \cap \text{cl}(\mathcal{U})) \geq \begin{cases} \bar{r}_+ - \bar{R} & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+ \\ \bar{R} - \bar{r}_- & \text{if } (\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^- \end{cases} \quad (3.12)$$

where $\bar{r}_\pm$ are the boundary radii (zeros) of the profile defining $(\bar{\mathcal{U}}, \bar{u})$ in (2.3)–(2.4). Furthermore, equality in (3.12) holds if, and only if $(\mathcal{U}, u) \equiv (\bar{\mathcal{U}}, \bar{u})$.

The proof is identical to [?, Theorem 5.1], so we omit it. To obtain the normalized estimate in Theorem C, we also need a pointwise lower bound for u in terms of the distance to $\partial\Omega$, obtained by comparison with the Serrin's ball profile as in [?, Lemma 2.1].

Lemma 3.7. *Let (Ω, u) solve (1.2) with $f(x) \leq f_k(x) = nkx + 1$. Suppose that Ω is contained in a convex domain of $\mathcal{M}$, and define $d(p) = \text{dist}(p, \partial\Omega)$ for all $p \in \text{cl}(\Omega)$. Then*

$$u(p) \geq \frac{2}{n} \cdot \frac{s_k\left(\frac{d(p)}{2}\right)^2}{s'_k(d(p))}, \quad \forall p \in \text{cl}(\Omega). \quad (3.13)$$

Moreover, if equality holds at some point $p \in \Omega$, then u attains its unique maximum at p , Ω is a geodesic ball centered at p , u depends only on the distance to p , and $f(x) = f_k(x)$.

Proof. Fix $p \in \Omega$, and write $r_p(q) = \text{dist}(p, q)$ for $q \in \text{cl}(\Omega)$. Under $\text{Ric} \geq (n-1)k$ g one has

$$\Delta r_p(q) \leq (n-1) \cot_k(r_p(q)), \quad q \in \text{cl}(\Omega).$$

Let

$$W^R(r) = \frac{2 \left[s_k\left(\frac{R}{2}\right)^2 - s_k\left(\frac{r}{2}\right)^2 \right]}{n s'_k(r)}, \quad r \in [0, R],$$

be the Serrin's ball profile of radius R in $\mathbb{M}^n(k)$. Since $f(x) \leq nkx + 1$, the function $w^R := W^R \circ r_p$ is a subsolution of (1.2) on the geodesic ball $B_R(p)$. Taking $R = d(p)$ and using the maximum principle gives $u \geq w^{d(p)}$ in $B_{d(p)}(p)$, hence

$$u(p) \geq w^{d(p)}(p) = \frac{2}{n} \cdot \frac{s_k\left(\frac{d(p)}{2}\right)^2}{s'_k(d(p))}.$$

The rigidity statement follows from the strong maximum principle. $\square$

We denote by

$$r_\Omega = \max \{d(p) = \text{dist}(p, \partial\Omega) : p \in \Omega\}$$

the inradius of Ω . We can now prove Theorem C.

Proof of Theorem C. Let $o \in \Omega$ such that $r_\Omega = \text{dist}(o, \partial\Omega)$. Developing the inequality of Lemma 3.7 for $\frac{2}{n}u(o) + ku(o)^2$ and using the upper bound for u gives

$$\frac{\tan_k(r_\Omega)^2}{n^2} \leq \frac{2}{n}u(o) + ku(o)^2 \leq \frac{2}{n}M + kM^2.$$

Combining this bound with Proposition 3.6 produces the lower bound for the function $\frac{nd(p)}{\tan_k(r_\Omega)}$ that yields the equation (1.5).

For rigidity, equality in (1.5) implies equality in (3.13), hence $f(x) = nkx + 1$, Ω is a geodesic ball centered at o , u depends only on r_o , and $\text{Max}(u) = \{o\}$. In particular, we get that equality can hold only if $(\bar{\mathcal{U}}, \bar{u}) \in \text{Model}_{M,k,f}^+$. Since $\Omega \setminus \text{Max}(u)$ is connected, equality in (3.12) also holds and the rigidity statement in Theorem A yields the warped product decomposition

$$g = d\Psi^2 + s_k(\Psi)^2 \bar{g}$$

on Ω , where Ψ is as in Definition 2.12 and $\bar{g}$ is the metric induced on Γ_0 . Since then $\tau(\Omega) = 1$, the conclusion follows from Remark 3.1. $\square$

A Appendix: ODE toolkit

In this appendix we collect and prove the ODE statements used in Section 2. Since most of the arguments are standard and already considered in [?], we only sketch the proofs and refer to our previous paper when necessary.

Assume that $f \in \mathcal{C}(\mathbb{R})$ satisfies the Standard Conditions. We study the equation

$$U''(r) + (n-1) \cot_k(r) U'(r) + f(U(r)) = 0, \quad r \in (0, \bar{r}_k), \quad (\text{A.1})$$

together with the initial conditions

$$U(R) = M > 0, \quad U'(R) = 0, \quad R \in [0, 2\bar{r}_k), \quad (\text{A.2})$$

with $\bar{r}_k = +\infty$ if $k \leq 0$ and $\bar{r}_k = \pi/\sqrt{k}$ if $k > 0$.

We begin by checking the monotonicity properties of $U_{R,M,k,f}$. In particular, we prove that f is k -admissible under the standard assumptions.

Proof of Proposition 2.4. Given any $R \in [0, \bar{r}_k]$ and $M \in \mathcal{I}_f$, the existence of a solution $U_{R,M,k,f}$ to (A.1)–(A.2) follows by standard ODE theory, so we omit the details. Let (e_1, e_2) be the maximal interval of definition of $U_{R,M,k,f}$. From the implicit representation

$$U_{R,M,k,f}(r) = M - \int_R^r \frac{1}{s_k(y)^{n-1}} \left(\int_R^y s_k(x)^{n-1} f(U_{R,M,k,f}(x)) dx \right) dy,$$

we obtain that, whenever $U_{R,M,k,f}(r)$ is positive, one has $U'_{R,M,k,f}(r) > 0$ for $r < R$ and $U'_{R,M,k,f}(r) < 0$ for $r > R$.

First, when $k > 0$ the existence of $r_-(R, M, k, f)$ and $r_+(R, M, k, f)$ follows by the same arguments as in [?, Theorem 4.1, item (2)]. Now, if $R \in \{0, \bar{r}_k\}$, *i.e.* a singular case. We proceed proof only the case $R = 0$, since the case $R = \bar{r}_k$ is similar. We proof with a standard fixed point argument, as in [?, ?]. As in [?, Subsection 3.2.1], for a non-negative integer k and $\varepsilon \in (0, \bar{r}_k)$, we denote by $\mathcal{C}_e^k(\varepsilon)$ the space of even $\mathcal{C}^k$ -functions on $[-\varepsilon, \varepsilon]$, equipped with the norm

$$\|Z\|_{k,\varepsilon} = \sum_{i=1}^k \sup_{s \in [-\varepsilon, \varepsilon]} |Z^{(i)}(s)|, \quad \forall Z \in \mathcal{C}_e^k(\varepsilon).$$

We also define $\mathcal{C}_e(\varepsilon) := \mathcal{C}_e^0(\varepsilon)$. For any $g \in \mathcal{C}_e(\varepsilon)$, define

$$A(f)(s) = M - \int_0^s \mu(y)^{-1} \left(\int_0^y \mu(x) f(x) dx \right) dy, \quad \forall s \in (-\varepsilon, \varepsilon), \quad (\text{A.3})$$

and set $A(f)(0) = 0$. It is straightforward to check that $A(f)$ is $\mathcal{C}^2$ on $(-\varepsilon, \varepsilon)$.

Define the operator $\mathcal{T} : \mathcal{C}_e^1(\varepsilon) \rightarrow \mathcal{C}_e^1(\varepsilon)$ by

$$\mathcal{T}(Z) = A(f(M + Z)).$$

Taking into account that

$$\cot(s) = \frac{1}{s} + O(s) \quad \text{and} \quad \sin(s) = s + O(s^3)$$

near 0, we check that there exist constants $C_1, C_2 > 0$ such that

$$C_1 s^{n-1} \leq s_k(r)^{n-1} \leq C_2 s^{n-1}$$

for s close to 0. In particular, there exists $C_3 > 0$ such that

$$\|\mathcal{T}(U)\|_{0,\varepsilon} \leq C_3 \varepsilon^2 \sup_{[M-\|U\|_{0,\varepsilon}, M+\|U\|_{0,\varepsilon}]} |f|.$$

Thus, if ε is small enough, we obtain $\mathcal{T}(B) \subsetneq B$, where $B \subset \mathcal{C}_e(\varepsilon)$ is the closed ball of radius 1 centered at 0. Consequently, by Schauder's fixed point theorem, $\mathcal{T}$ has a fixed point $U_{0,M,f} \in B$, which is a weak solution to (2.3)–(2.4). By construction, $U_{0,M,f}$ is $\mathcal{C}^2$ on $(-\varepsilon, \varepsilon)$, hence it is a classical solution there. The existence of $r_+(0, M, f)$ and the stated qualitative properties follow as in the nonsingular case.

Now we assume that $k \leq 0$. Since the argument is the same in both cases, we only present the proof when $k < 0$.

The existence of $r_-(R, M, k, f)$ and $r_+(R, M, k, f)$ when $e_2 < +\infty$ is immediate from the boundary behavior of solutions to differential equations with Lipschitz coefficients. Thus, we only prove the existence of $r_+(R, M, k, f) > R$ when $e_2 = +\infty$.

Since f satisfies the Standard Conditions, suppose that $f(x) \geq -nx + 1$. Given a fixed point $p \in \mathbb{H}^n(k)$, define the function $r_p = \text{dist}_{\mathbb{H}^n(k)}(p, \cdot)$. Then set $\tilde{u} = U_{R,M,k,f} \circ r_p$ inside the set $\mathcal{U} := \{r_p(q) > R\}$. Note that $\tilde{u}$ would be a piece of a model solution in $\mathbb{H}^n(k)$ if there exist $r_-(R, M, k, f) \leq R < r_+(R, M, k, f)$ as in Definition 2.3. Take $q \in \{r_p = R\}$. There exists a function v and a geodesic ball $B \subset \mathbb{H}^n(k)$ centered at q such that v is positive and solves $\Delta v - nv + 1 = 0$ in B , and satisfies $v = 0$ on ∂B .

Arguing by contradiction, suppose that $\tilde{u}$ is positive in all of $\mathcal{U}$. Then there exists $\alpha_0 > 0$ such that $\alpha_0 v < \tilde{u}$ in B . By increasing α , we obtain $\alpha_1 > \alpha_0$ such that the graphs of $\alpha_1 v$ and $\tilde{u}$ touch tangentially for the first time. But then, since $f(x) \geq -nx + 1$, the maximum principle implies $\tilde{u} \equiv \alpha_1 v$ in B , which is a contradiction. □

Proposition A.1. *Let $f(x) = f_k(x) = nkx + 1$. Then for the corresponding solution of (2.3)–(2.4), one has*

$$\lim_{R \rightarrow 0^+} \bar{\tau}_{M,k,f}^-(R) = +\infty.$$

Proof. Denote by $V_{R,M,k}$ the solution to (2.3)–(2.4) with $f = f_k$. For the case $k = 0$ the behavior of $\bar{\tau}_{M,k,f}^-(R)$ is proved in [?, ?]. Thus, we focus on the case $k \neq 0$. We shall assume

that $k = \pm 1$, since otherwise we can work with $\tilde{V}_{R,M,k}(r) = |k| V_{R,M,k}(r/\sqrt{|k|})$. In this case, the function $V_{R,M,k}$ can be expressed as

$$V_{R,M,k}(r) = M + c_k(r) \left(A(R, M, k) + B(R, M, k) \left(\frac{-k}{s_k(r)} + G_k(r) \right) \right),$$

for some positive numbers $A(R, M, k)$ and $B(R, M, k)$ and

$$G_k(r) := \begin{cases} \int_0^{\cosh(r)} \frac{1 - (1 - t^2)^{\frac{n}{2}}}{t^2(1 - t^2)^{\frac{n}{2}}} dt, & k = -1, \\ \int_0^{\cos(r)} \frac{1 - (t^2 - 1)^{\frac{n}{2}}}{t^2(t^2 - 1)^{\frac{n}{2}}} dt, & k = 1. \end{cases}$$

Since $G_k(r) \approx r^{1-n}$ near $r = 0$, we deduce that $V'_{R,M,k}(r_{\pm})$ diverges as $R \rightarrow 0^+$, and therefore $\bar{\tau}_{M,k,f}^-(R) \rightarrow +\infty$. $\square$

Proof of Proposition 2.7. First we assume $f(0) > 0$ also when $k > 0$, and by scaling we may fix $f(0) = 1$. Suppose for contradiction that

$$\lim_{R \rightarrow 0^+} U'_{R,M,k}(r_-(R, M, k)) = \alpha < +\infty.$$

Let $V_{R,M,k}$ be the Serrin's solution for $f_k(x) = nkx + 1$. By Proposition A.1, for sufficiently small R we have $V'_{R,M,k}(r_-(R, M, k)) > \alpha$. Since $f(x) \geq f_k(x)$, a comparison of $U_{R,M,k,f}$ and $V_{R,M,k}$ on their common domain (via the maximum principle) yields a contradiction. Hence $\bar{\tau}_{M,k,f}^-(R) \rightarrow +\infty$ as $R \rightarrow 0^+$.

For the second claim, if $k > 0$ then the symmetry relation in Proposition 2.4 gives $\bar{\tau}_{M,k,f}^+(\bar{r}_k) = \bar{\tau}_{M,k,f}^-(\bar{r}_k)$, so $\tau_{k,f}^+(M) \geq \tau_{k,f}^-(M)$. If $k \leq 0$, observe that $\bar{\tau}_{M,k,f}^+(0) = 1$ while $\bar{\tau}_{M,k,f}^-(0^+) = +\infty$, so $\bar{\tau}_{M,k,f}^+(R) < \bar{\tau}_{M,k,f}^-(R)$ for small $R > 0$. If there were some $R > 0$ with $\bar{\tau}_{M,k,f}^+(R) = \bar{\tau}_{M,k,f}^-(R)$, then the corresponding model solution would have constant Neumann data on the boundary, and thus (by Serrin's theorem [?, ?]) $\Omega_{R,M,k,f}$ would be a geodesic ball. But this can only occur if $R = 0$, a contradiction. Therefore $\bar{\tau}_{M,k,f}^+ < \bar{\tau}_{M,k,f}^-$ for all $R > 0$, yielding $\tau_{k,f}^+(M) \leq \tau_{k,f}^-(M)$.

In the case $f(0) = 0$ when $k > 0$, the proposition follows with the same arguments using the function $V_{R,M,k} + \frac{1}{nk}$. $\square$

Now suppose further that $f \in \mathcal{C}^1(\mathbb{R})$ and there exists $\bar{M} > 0$ such that

$$f(x) \geq f'(x)x \quad \forall x \in (0, \bar{M}) \subset \mathcal{I}_f.$$

In this case, we can prove the monotonicity properties of the functions $\bar{\tau}_{M,k,f}^{\pm}$ defined in (2.7), as stated in Proposition 2.8. The key observation is that the function $H_{R,k,f} := \partial_R(U_{R,M,k,f})$ satisfies the differential equation

$$H''(r) + (n-1) \cot_k(r) H'(r) + f'(U(r)) H(r) = 0, \quad \forall r \in [\bar{r}_-, \bar{r}_+],$$

where we write $\bar{r}_\pm = r_\pm(R, M, k, f)$. Hence, we can follow the proof of item (4) in [?, Theorem 4.1].

Proof of Proposition 2.8. To obtain the monotonicity properties of the functions $\bar{\tau}_{M,k,f}^\pm$, it is enough to study the sign of the functions

$$\varphi_{M,k,f}^\pm(R) := \partial_R(U_{R,M,k,f}(\bar{r}_\pm)^2) = -2 \left(U_{R,M,k,f}''(\bar{r}_\pm) H_{R,k,f}(\bar{r}_\pm) - H_{R,k,f}'(\bar{r}_\pm) U_{R,M,k,f}'(\bar{r}_\pm) \right).$$

Proceeding as in the proofs of Claims A and B in [?, Theorem 4.1], using the Killing vector field

$$\tilde{Y} = \cos \theta_{n-1} \prod_{j=1}^{n-2} \sin \theta_j \partial_r + \cot_k(r) \left(\cos \theta_{n-1} \sum_{i=1}^{n-2} \frac{\prod_{j=1}^{n-2} \sin \theta_j}{\prod_{j=1}^{i-1} \sin \theta_j} \partial_{\theta_i} - \frac{\sin \theta_{n-1}}{\prod_{j=1}^{n-2} \sin \theta_j} \partial_{\theta_{n-1}} \right)$$

in $\mathbb{M}^n(k)$, we obtain that $\varphi_{M,k,f}^-$ and $\varphi_{M,k,f}^+$ are negative and positive, respectively, on $[0, \bar{r}_k)$. In particular, $\bar{\tau}_{M,k,f}^-$ and $\bar{\tau}_{M,k,f}^+$ are strictly decreasing and strictly increasing, respectively. Note that $\bar{\tau}_{M,k,f}^+(0) = 1$ by definition, and it is clear that

$$\lim_{R \rightarrow 0^+} \bar{\tau}_{M,k,f}^-(R) = +\infty.$$

On the other hand, one has

$$\lim_{R \rightarrow \bar{r}_k} \bar{\tau}_{M,k,f}^\pm(R) = \tau_{k,f}^\pm(M) > 0,$$

and the implicit function theorem implies that these quantities depend $\mathcal{C}^1$ on M .

In the case $k > 0$, the symmetry of the functions $U_{R,M,k,f}$ with respect to $\bar{r}_k/2$ implies that $\bar{\tau}_{M,k,f}^-(\bar{r}_k) = \bar{\tau}_{M,k,f}^+(\bar{r}_k)$. $\square$

B Appendix: Gradient Estimates

In this section we present an elliptic inequality involving the functions

$$W = |\nabla u|^2 \quad \text{and} \quad \bar{W} = |\nabla \bar{u}|^2 \circ \Psi \quad \text{in } \mathcal{U},$$

used in the proof of Theorem A. In particular, we prove the following result.

Lemma B.1. *Let*

$$F_\beta = \left(\frac{s_k(\Psi)}{\bar{U}'(\Psi)} \right)^{2\frac{n-1}{n}} (W - \bar{W}) \quad \text{in } \mathcal{U}. \quad (\text{B.1})$$

Then F_β satisfies the following elliptic inequality in $\mathcal{U}$:

$$\Delta F_\beta - 2 \frac{n-1}{n} \lambda(\Psi) \langle \nabla F_\beta, \nabla u \rangle - 2 \frac{n-1}{n} \frac{|\nabla u|^2}{\bar{U}'(\Psi)^2} \mu(\Psi) F_\beta \geq 0, \quad (\text{B.2})$$

where we define

$$\lambda(r) := \frac{-\bar{U}''(r) + \cot_k(r) \bar{U}'(r)}{\bar{U}'(r)^2}, \quad \forall r \in [\bar{r}_-, \bar{R}) \cup (\bar{R}, \bar{r}_+], \quad (\text{B.3})$$

and

$$\mu(r) := f'(\bar{U}(r)) - nk + \frac{n+2}{n} \lambda(r) f(\bar{U}(r)). \quad (\text{B.4})$$

Then $\mu(r)$ is non-negative in $(\bar{r}_-, \bar{R}) \cup (\bar{R}, \bar{r}_+)$.

Proof. From now on, following the notation of [?], we denote $\chi_\pm = \chi$ and carry out the computations in a unified way for both monotonicity branches. We also denote derivatives with respect to u by a dot and derivatives with respect to Ψ by $'$.

The proof follows as in [?, Theorem 3.5]. The idea is to derive an elliptic inequality involving the functions W and $\bar{W}$ defined in Section 3. We recall that

$$W = |\nabla u|^2 \quad \text{and} \quad \bar{W} = |\nabla \bar{u}|^2 \circ \Psi \quad \text{in } \mathcal{U},$$

where $\Psi : \mathcal{U} \rightarrow (\bar{r}_-, \bar{R}) \cup (\bar{R}, \bar{r}_+)$ is the pseudo-radial function defined in Definition 2.12. We also recall that $\bar{U}$ satisfies

$$\bar{U}''(r) + (n-1) \cot_k(r) \bar{U}'(r) + f(\bar{U}) = 0. \quad (\text{B.5})$$

From the definition of Ψ , we have $u = \bar{U} \circ \Psi$ on $\mathcal{U}$. Recall that $F_\beta = \beta \cdot (W - \bar{W})$, where

$$\beta(p) = \left(\frac{s_k(\Psi(p))}{\bar{U}'(\Psi(p))} \right)^{2 \frac{n-1}{n}}, \quad \forall p \in \mathcal{U}.$$

Using that $\text{Ric} \geq (n-1)kg$, it follows from the Bochner formula that

$$\frac{1}{2} \Delta |\nabla u|^2 \geq |\nabla^2 u|^2 + ((n-1)k - f'(u)) |\nabla u|^2.$$

Then, by expanding the squared norm

$$\left| \nabla^2 u + \lambda(\Psi) du \otimes du + \frac{1}{n} (f(u) - \lambda(\Psi) |\nabla u|^2) g \right|^2 \geq 0, \quad (\text{B.6})$$

we obtain, after simplification, the inequality

$$\Delta F_\beta - 2 \frac{n-1}{n} \lambda(\Psi) \langle \nabla F_\beta, \nabla u \rangle - 2 \frac{n-1}{n} \frac{|\nabla u|^2}{\bar{U}'(\Psi)^2} \mu(\Psi) F_\beta \geq 0$$

in $\mathcal{U}$. Moreover, equality at some point forces equality in (B.6). Thus, to ensure the inequality has the correct sign structure for the maximum principle, it remains to show that $\mu \geq 0$ in $(\bar{r}_-, \bar{R}) \cup (\bar{R}, \bar{r}_+)$ under hypothesis (2.13).

First, note that up to scaling the solution and the metric, we may restrict to the cases $k = -1, 0$, or 1 . Observe that when $k = -1, 0$ and $r \in (\bar{r}_-, \bar{R})$, or when $k = 1$ and $r \in (\bar{r}_-, \bar{R})$, or when $k = 1$ and $r \in [\bar{r}_1/2, \bar{r}_+)$ provided $\bar{r}_+ > \bar{r}_1/2$, one can check directly that $\mu(r) > 0$ using (2.3). For the remaining cases, consider the change of variable

$$r(t) = \begin{cases} \operatorname{arccosh}(t) & \text{if } k = -1 \\ \sqrt{2t} & \text{if } k = 0, \\ \arccos(t) & \text{if } k = 1, \end{cases}$$

and define $\bar{V}(t) := \bar{U}(r(t))$. A direct computation shows that

$$\mu(t) = \frac{1}{r'(t)^2} \left(\frac{n+2}{n} \bar{V}''(t)^2 - \bar{V}'''(t) \bar{V}'(t) \right), \quad \forall t \in (\bar{R}, \bar{r}_+).$$

The idea is to argue as in item (6) in [?, Theorem 4.1]. Since the three cases $k = -1, 0, 1$ are very similar, we only write the proof for $k = 0$.

Define $\bar{T} = \bar{R}^2/2$ and $\bar{t}_+ = \bar{r}_+^2/2$. Using that $\bar{U}$ solves (2.3), we obtain that $\bar{V}$ solves

$$2t \bar{V}''(t) + n \bar{V}'(t) + f(\bar{V}) = 0,$$

and thus $\bar{V}'$ solves

$$2t \bar{V}'''(t) + (n+2) \bar{V}''(t) + f'(\bar{V}) \bar{V}'(t) = 0.$$

Integrating this equation between $\bar{T}$ and t , we obtain

$$\bar{V}''(t) = -\frac{1}{(2t)^{\frac{n+2}{2}}} \left((2\bar{T})^{\frac{n}{2}} f(M) + \int_{\bar{T}}^t (2s)^{\frac{n}{2}} f'(\bar{V}(s)) \bar{V}'(s) ds \right),$$

where $M = u_{\max}$. Note that $f' \geq 0$ and $\bar{V}'(t) < 0$ for $t \in (\bar{T}, \bar{t}_+)$, so the integral term is non-positive on this interval. Now take $\bar{T} < a < b$. Then

$$\begin{aligned} \bar{V}''(a) - \bar{V}''(b) &= \left(\frac{1}{(2b)^{\frac{n+2}{2}}} - \frac{1}{(2a)^{\frac{n+2}{2}}} \right) \left((2\bar{T})^{\frac{n}{2}} f(M) - \int_{\bar{T}}^a (2s)^{\frac{n}{2}} f'(\bar{V}(s)) \bar{V}'(s) ds \right) \\ &\quad + \frac{1}{(2b)^{\frac{n+2}{2}}} \int_a^b (2s)^{\frac{n}{2}} f'(\bar{V}(s)) \bar{V}'(s) ds \leq 0. \end{aligned}$$

This implies that $\bar{V}''' \geq 0$ on $(\bar{T}, \bar{t}_+)$, hence $\mu \geq 0$ on this interval. This concludes the proof. $\square$

Data availability

The code for the graphs are available in the following GitHub repository <https://github.com/FernanGI/ComparisonTechniquesLowerBound>.

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